Design of a Two-Stage CMOS Op Amp

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Abstract

A two-stage Miller-compensated CMOS op amp is designed in the GF180MCU 3.3 V process [3] using an overdrive-voltage-based sizing method. Without compensation, the amplifier is unstable, with a phase margin of $-8.2°$. A 350 fF compensation capacitor is used, chosen so that the first stage, not the output stage, limits the slew rate. The capacitor also makes one pole more dominant, through pole splitting, leading to a more stable frequency response. The right-half-plane zero it creates is moved to infinity or into the left half-plane using a series resistance. This is implemented in two ways: using a resistor, and using an NMOS transistor in the triode region (M_Z). At TT and 27 °C, the final design with M_Z achieves a DC gain of 81.7 dB, a unity-gain frequency of 34.9 MHz, a phase margin of 64.9° and a gain margin of 32.4 dB, with a power consumption of 0.427 mW.

Across all process corners and temperatures from -40 to 125 °C, the phase margin with M_Z stays between 64.2° and 66.8°, compared with 61.6° to 71.6° for a PDK poly resistor including its process corners and temperature coefficient, because M_Z holds the phase margin better over temperature.

Monte Carlo analysis shows the opposite trade-off from chip to chip: a PDK poly resistor gives a phase-margin standard deviation of 0.58°, against 1.81° for M_Z. The input offset has a standard deviation of about 8.96 mV for the triode implementation and 8.74 mV for the PDK poly resistor. The larger chip-to-chip spread of M_Z comes from its narrow bias devices, which fall in a different model bin from M_6 and M_Z; widening them into the same bin reduces the phase-margin standard deviation to 0.57°, matching the poly resistor, at the cost of 187 μW.

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1 Introduction

Op amps are of considerable importance in IC design. Their ideal qualities, such as high input resistance and high open-loop gain, combined with their use in negative feedback designs such as inverting amplifiers and buffers, prove to be incredibly useful in a variety of designs. The two-stage CMOS op amp is just one popular design.

This report describes the design of a two-stage op-amp in the GF180MCU 3.3 V process. The design targets a phase margin of at least 60° with a 1 pF load.

The design is simulated in LTspice using the GF180MCU device models. The corner, temperature and Monte Carlo analyses are run in ngspice, which reads the corner and statistical sections of the PDK model files.

Section 2 describes the circuit and Sections 3 and 4 derive the transistor sizes and bias currents. Sections 5 to 7 present the frequency response and the large-signal step response. Section 8 summarizes the final design, its frequency response, its slew rate, its CMRR, and its power consumption. Section 9 examines its performance across process corners, temperature, and device mismatch.

2 Circuit Description

A schematic of the two-stage CMOS op amp is shown in Fig. 1. Transistors M_1 and M_2 implement the PMOS input differential pair, with M_3 and M_4 implementing the NMOS current-mirror load. I_{REF} is the reference current. M_8 and M_5 form a current mirror to bias M_1 and M_2. This implements the first stage of the op amp. For the second stage, we have M_6 in a common-source configuration, with M_7 as a PMOS load, which also forms a current mirror with M_8 and M_5. Compensation capacitor C_{C} is connected between the outputs of the first and second stages. C_{L} represents the capacitive load.

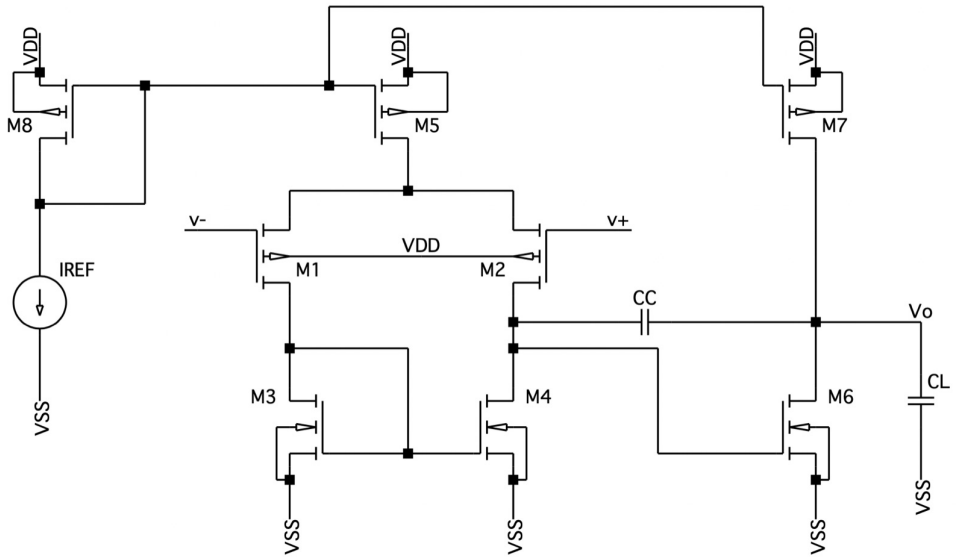


Figure 1: Two-Stage CMOS Op Amp Schematic

3 Design Methodology

For the design, a common overdrive voltage V_{OV} is used for all transistors. As a result, using the square-law relationship, each transistor's DC current is proportional to its W/L. We can relate the currents as a simple ratio of their W/L's. To eliminate or reduce systematic offset, the following condition must be satisfied [1]:

$$\frac{(W/L)_6}{(W/L)_4} = 2\,\frac{(W/L)_7}{(W/L)_5} \tag{1}$$

We target an overdrive voltage V_{OV} of 0.2 V. We use a common length of L=0.56 μm for all transistors.

4 DC Bias Point Design

4.1 Sizing

In this design, we select $I_{\mathrm{REF}} = 20\,\mu\mathrm{A}$. We also select $I_5 = 20\,\mu\mathrm{A}$ to bias M_5. M_1, M_2, M_3 and M_4 will each be biased by $\frac{I_5}{2} = 10\,\mu\mathrm{A}$.

PMOS Sizing Using $V_{\mathrm{SG}} = 0.9\,\mathrm{V}$, and $V_{SD} = V_{DD}/3 = 1.1\,\mathrm{V}$, we perform a DC operating point simulation for the PMOS with $L = 0.56\,\mu\mathrm{m}$ and $W = 1\,\mu\mathrm{m}$. From the simulation, we have that $|V_{\mathrm{th}}| = 0.776\,\mathrm{V}$ and $|V_{\mathrm{dsat}}| = 0.146\,\mathrm{V}$. Determining the V_{SG} at which $V_{\mathrm{OV}} = 0.2\,\mathrm{V}$, we have

$$|V_{\mathrm{OV}}| = V_{\mathrm{SG}} - |V_{\mathrm{th}}| \;\Rightarrow\; V_{\mathrm{SG}} = |V_{\mathrm{OV}}| + |V_{\mathrm{th}}| = 0.2\,\mathrm{V} + 0.776\,\mathrm{V} = 0.976\,\mathrm{V}.$$

Using $V_{\mathrm{SG}} = 0.976\,\mathrm{V}$ and $V_{\mathrm{SD}} = 1.1\,\mathrm{V}$, we have that $|V_{\mathrm{th}}| = 0.776\,\mathrm{V}$ and $|V_{\mathrm{dsat}}| = 0.203\,\mathrm{V}$, which is sufficiently close to the target $|V_{\mathrm{OV}}|$ of 0.2 V. At $|V_{\mathrm{SG}}| = 0.976\,\mathrm{V}$ and $|V_{\mathrm{SD}}| = 1.1\,\mathrm{V}$, $|I_{\mathrm{D}}| = 1.46\,\mu\mathrm{A}$. Thus,

$$\frac{I_{\mathrm{D}}}{W} = \frac{1.46\,\mu\mathrm{A}}{1\,\mu\mathrm{m}} = 1.46\,\frac{\mu\mathrm{A}}{\mu\mathrm{m}}$$

NMOS Sizing Using $V_{\mathrm{GS}} = 0.9\,\mathrm{V}$ and $V_{DS} = V_{DD}/3 = 1.1\,\mathrm{V}$, we perform a DC operating point simulation for the NMOS with $L = 0.56\,\mu\mathrm{m}$ and $W = 1\,\mu\mathrm{m}$. From the simulation, we have that $|V_{\mathrm{th}}| = 0.691\,\mathrm{V}$ and $|V_{\mathrm{dsat}}| = 0.223\,\mathrm{V}$. Determining the V_{GS} at which $V_{\mathrm{OV}} = 0.2\,\mathrm{V}$, we have

$$|V_{\mathrm{OV}}| = V_{\mathrm{GS}} - |V_{\mathrm{th}}| \;\Rightarrow\; V_{\mathrm{GS}} = |V_{\mathrm{OV}}| + |V_{\mathrm{th}}| = 0.2\,\mathrm{V} + 0.691\,\mathrm{V} = 0.891\,\mathrm{V},$$

using $V_{\mathrm{GS}} = 0.891\,\mathrm{V}$ and $V_{\mathrm{DS}} = 1.1\,\mathrm{V}$, we have that $|V_{\mathrm{th}}| = 0.691\,\mathrm{V}$ and $|V_{\mathrm{dsat}}| = 0.216\,\mathrm{V}$, sufficiently close to the target V_{OV} of 0.2 V. At $V_{\mathrm{GS}} = 0.891\,\mathrm{V}$ and $V_{\mathrm{DS}} = 1.1\,\mathrm{V}$, $I_{\mathrm{D}} = 5.95\,\mu\mathrm{A}$. Thus,

$$\frac{I_{\mathrm{D}}}{W} = \frac{5.95\,\mu\mathrm{A}}{1\,\mu\mathrm{m}} = 5.95\,\frac{\mu\mathrm{A}}{\mu\mathrm{m}}$$

Sizing M_1 and M_2 For the PMOS devices, since $I_{\mathrm{M1}} = I_{\mathrm{M2}} = 10\,\mu\mathrm{A}$, using $\frac{I_{\mathrm{D}}}{W} = 1.46\,\mu\mathrm{A}/\mu\mathrm{m}$, we have

$$W_1 = W_2 = \frac{I_{\mathrm{D}}}{1.46\,\mu\mathrm{A}/\mu\mathrm{m}} = \frac{10\,\mu\mathrm{A}}{1.46\,\mu\mathrm{A}/\mu\mathrm{m}} = 6.85\,\mu\mathrm{m}$$

Sizing M_3 and M_4 For the NMOS devices, since $I_{\mathrm{M3}} = I_{\mathrm{M4}} = 10\,\mu\mathrm{A}$, using $\frac{I_{\mathrm{D}}}{W} = 5.95\,\mu\mathrm{A}/\mu\mathrm{m}$, we have

$$W_3 = W_4 = \frac{I_{\mathrm{D}}}{5.95\,\mu\mathrm{A}/\mu\mathrm{m}} = \frac{10\,\mu\mathrm{A}}{5.95\,\mu\mathrm{A}/\mu\mathrm{m}} = 1.68\,\mu\mathrm{m}$$

Sizing M_5 For the PMOS device, since $I = I_{M5} = 20\,\mu A$, using $\frac{I_D}{W} = 1.46\,\mu A/\mu m$, we have

$$W_5 = \frac{I_D}{1.46\,\mu A/\mu m} = \frac{20\,\mu A}{1.46\,\mu A/\mu m} = 13.7\,\mu m$$

Sizing M_6 and M_7 To size M6 and M7, Eq. (1) needs to be satisfied. For M7 and M6, we select $I_{M7} = I_{M6} = 80\,\mu A$. For the PMOS devices, using $I_D/W = 1.46\,\mu A/\mu m$, we have

$$W_7 = \frac{I_D}{1.46\,\mu A/\mu m} = \frac{80\,\mu A}{1.46\,\mu A/\mu m} = 54.8\,\mu m$$

Using Equation (1), we have

$$\frac{(W/L)_6}{(W/L)_4} = 2\,\frac{(W/L)_7}{(W/L)_5}$$

Since we have the same length for all transistors,

$$\frac{W_6}{W_4} = 2\,\frac{W_7}{W_5} \;\Rightarrow\; W_6 = 2\,\frac{W_7}{W_5}\,W_4 = 2\,\frac{54.8\,\mu m}{13.7\,\mu m}\,(1.68\,\mu m) = 13.44\,\mu m$$

Sizing M_8 We select $I_{M8} = I_{M5} = 20\,\mu A$. Thus, $W_5 = W_8 = 13.7\,\mu m$.

A schematic of the two-stage CMOS op amp with sizing and DC components is shown in Fig. 2.

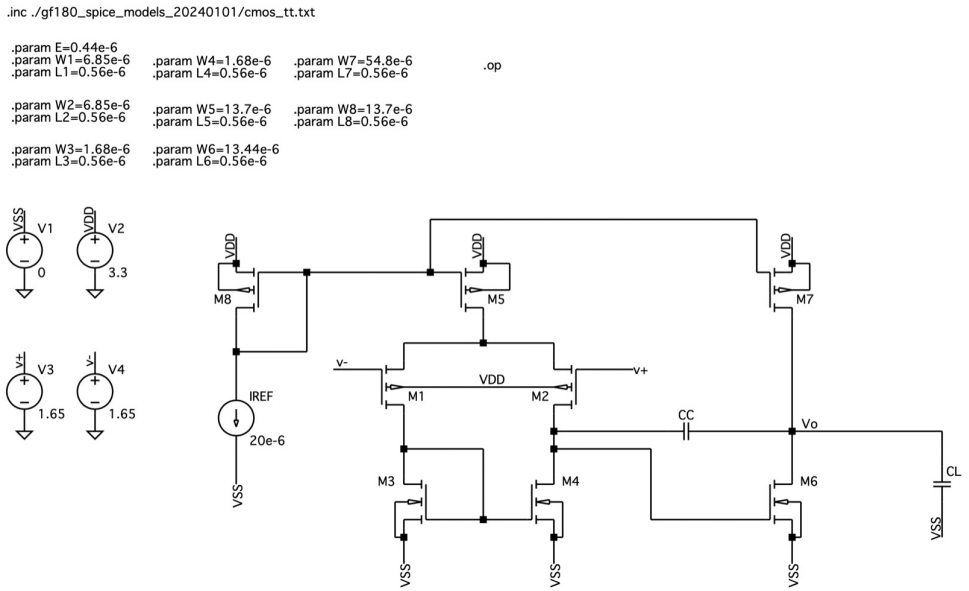


Figure 2: Two-Stage CMOS Op Amp Schematic With Sizing

4.2 DC Operating Point

A summary of the sizing is provided in Table 1:

Table 1: Transistor sizing summary

Transistor	Type	Width (μm)	Length (μm)
M1	PMOS	6.85	0.56
M2	PMOS	6.85	0.56
M3	NMOS	1.68	0.56
M4	NMOS	1.68	0.56
M5	PMOS	13.7	0.56
M6	NMOS	13.44	0.56
M7	PMOS	54.8	0.56
M8	PMOS	13.7	0.56

A summary of the DC operating point is provided in Table 2:

Table 2: DC operating point parameters

Transistor	M1	M2	M3	M4	M5	M6	M7	M8
I_{D} (μA)	9.73	9.73	9.73	9.73	19.5	81.9	81.9	20
$\lvert V_{\mathrm{GS}}\rvert$ (V)	1.17	1.17	0.889	0.889	0.98	0.889	0.98	0.98
$\lvert V_{\mathrm{DS}}\rvert$ (V)	1.93	1.93	0.889	0.889	0.483	1.62	1.68	0.98
$\lvert V_{\mathrm{BS}}\rvert$ (V)	0.483	0.483	0	0	0	0	0	0
$\lvert V_{\mathrm{th}}\rvert$ (V)	0.983	0.983	0.699	0.699	0.79	0.708	0.79	0.79
$\lvert V_{\mathrm{dsat}}\rvert$ (V)	0.202	0.202	0.209	0.209	0.193	0.203	0.194	0.193
g_{m} (mS)	0.0865	0.0865	0.0798	0.0798	0.17	0.676	0.712	0.174
g_{mb} (mS)	0.0309	0.0309	0.0292	0.0292	0.0768	0.246	0.322	0.079
g_{ds} (μS)	0.274	0.274	0.572	0.572	1.5	3.23	2.38	0.847
C_{bd} (fF)	5.19	5.19	1.71	1.71	12.8	11.5	43.2	11.8
C_{bs} (fF)	6.46	6.46	2.13	2.13	14.5	16	57.5	14.5

4.3 Input Common-Mode Range and Output Swing

Input Common-Mode Range For $M_{1,2}$ to remain in saturation,

$$V_{DG1,2} \leq |V_{tp1,2}| \;\Rightarrow\; V_{D1,2} - V_{ICM} \leq |V_{tp1,2}|$$

and, since M_3 is diode-connected,

$$V_{D1,2} = V_{D3} = V_{G3,4} = V_{OV3,4} + V_{tn3,4} - V_{SS}$$

Thus,

$$V_{ICM} \geq V_{OV3,4} + V_{tn3,4} - |V_{tp1,2}| - V_{SS}$$

For M_5 to remain in saturation,

$$V_{SD5} = V_{DD} - V_{D5} \geq V_{OV5}$$

and

$$V_{D5} = V_{S1,2} = V_{ICM} + V_{OV1,2} + |V_{tp1,2}|$$

Thus,

$$V_{ICM} \leq V_{DD} - V_{OV5} - V_{OV1,2} - |V_{tp1,2}|$$

Therefore, the input common-mode range is

$$V_{OV3,4} + V_{tn3,4} - |V_{tp1,2}| - V_{SS} \;\le\; V_{ICM} \;\le\; V_{DD} - V_{OV5} - V_{OV1,2} - |V_{tp1,2}| \tag{2}$$

From Eq. (2) we have

$$-0.075\,\mathrm{V} \;\le\; V_{ICM} \;\le\; 1.922\,\mathrm{V}$$

Output Swing For M_7 to remain in saturation,

$$V_{SD7} = V_{DD} - V_o \geq V_{OV7}$$

Thus,

$$V_o \leq V_{DD} - V_{OV7}$$

For M_6 to remain in saturation,

$$V_{DS6} = V_o + V_{SS} \geq V_{OV6}$$

Thus,

$$V_o \geq V_{OV6} - V_{SS}$$

Therefore, the output swing is

$$V_{OV6} - V_{SS} \;\le\; V_o \;\le\; V_{DD} - V_{OV7} \tag{3}$$

From Eq. (3) we have

$$0.203\,\mathrm{V} \;\le\; V_o \;\le\; 3.11\,\mathrm{V}$$

5 Small-Signal AC Analysis

5.1 Hand Analysis

For the two-stage op-amp, the DC gain is [1]

$$A = A_{v1}A_{v2}$$

where

$$A_{v1} = g_{m1}\,(r_{o2} \parallel r_{o4}) = g_{m1}\,R_1$$

is the gain of the first stage and R_1 is the output resistance at the first stage, and

$$A_{v2} = g_{m6}\,(r_{o6} \parallel r_{o7}) = g_{m6}\,R_2$$

is the gain of the second stage and R_2 is the output resistance of the second stage. From Table 2, we have

$$R_1 = r_{o2} \parallel r_{o4} = \frac{1}{g_{ds2} + g_{ds4}} = \frac{1}{2.74 \times 10^{-7} + 5.72 \times 10^{-7}} = 1.182\,\mathrm{M\Omega}$$

$$R_2 = r_{o6} \parallel r_{o7} = \frac{1}{g_{ds6} + g_{ds7}} = \frac{1}{3.23 \times 10^{-6} + 2.38 \times 10^{-6}} = 178.3\,\mathrm{k\Omega}$$

$$A_{v1} = g_{m1}R_1 = (8.65 \times 10^{-5})(1.18 \times 10^{6}) = 102.2\,\mathrm{V/V}$$

$$A_{v2} = g_{m6}R_2 = (6.76 \times 10^{-4})(1.78 \times 10^{5}) = 120.5\,\mathrm{V/V}$$

Thus,

$$A = A_{v1}A_{v2} = (102.2\,\mathrm{V/V})(120.5\,\mathrm{V/V}) = 12315\,\mathrm{V/V} = 81.8\,\mathrm{dB}$$

6 Frequency Compensation

6.1 AC Response Without Compensation

Figure 3 shows the setup for AC analysis, with a load capacitor of 1 pF and the compensating capacitor removed. The capacitor C_{FB} and inductor L_{FB} are used to set the DC operating point of the op amp. The netlist can be found in Section B.1 with `CC` and `MZ` removed.

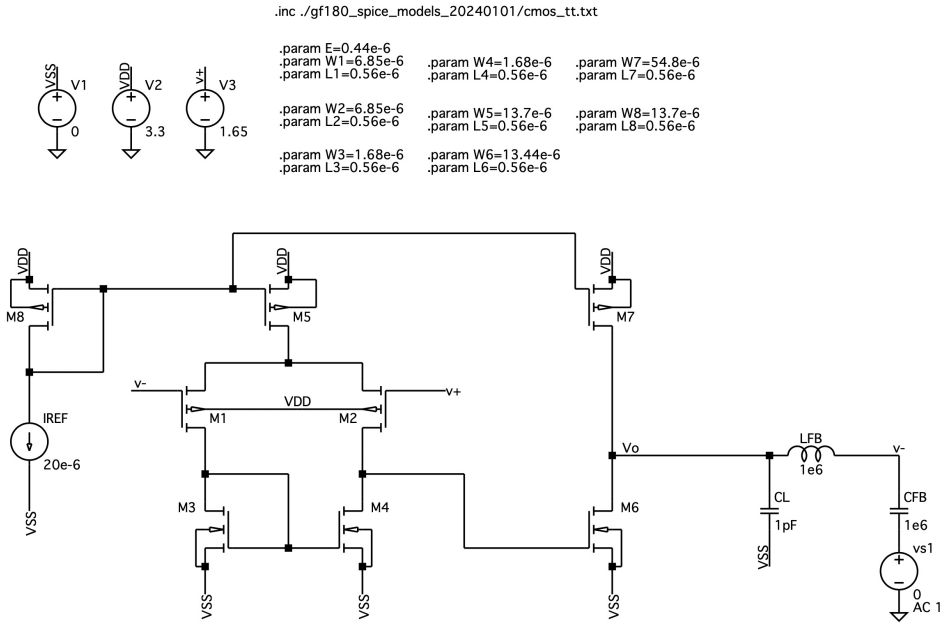


Figure 3: Open-loop Frequency Response Setup (Without Compensation Capacitor)

Performing the AC analysis, we have the following frequency response in Fig. 4

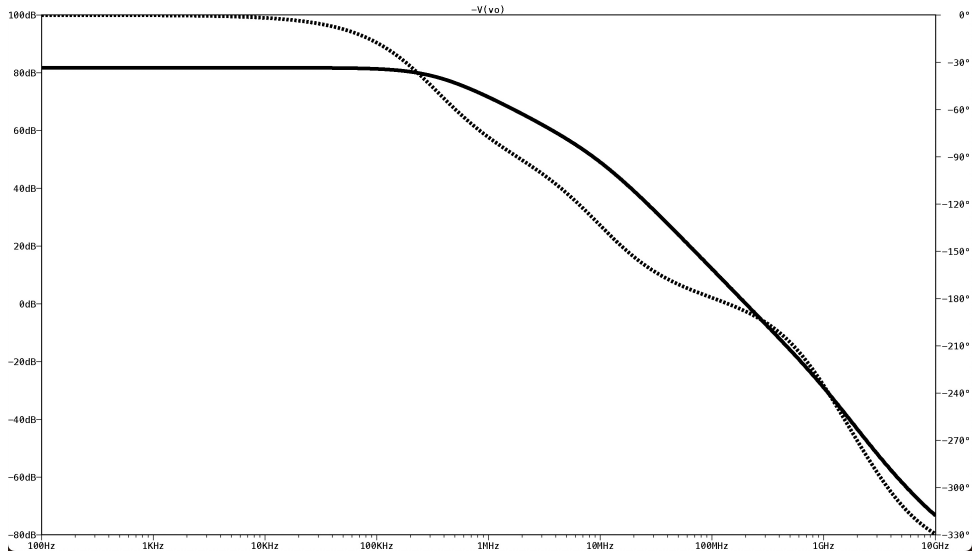


Figure 4: Open-loop Frequency Response (Without Compensation Capacitor)

The simulated DC gain is 81.74 dB, very close to the calculated DC gain. We find that f_t=201.2 MHz and the phase is approximately $-188.2°$. The -3 dB bandwidth is 326.5 kHz. Therefore, the phase margin is

$$\text{PM} = 180° - 188.2° = -8.2°$$

Since the PM $< 0°$, the op amp is unstable in unity-gain feedback. The use of the compensation capacitor alleviates this issue using pole splitting. Pole splitting makes one of the poles more dominant, while moving the other to a higher frequency. This allows for the AC response to have a uniform gain roll off at a rate of -20 dB/decade to 0 dB from the more dominant pole, increasing stability.

C_1 models the capacitance between the output node of the first stage and ground. C_2 models the capacitance between the output node of the second stage and ground. Approximating these capacitances we have,

$$C_1 \approx C_{\text{bd2}} + C_{\text{bd4}} + C_{\text{gs6}} + (1 + A_{v2})C_{\text{gd6}} \tag{4}$$

where $(1 + A_{v2})\, C_{\text{gd6}}$ is the Miller-multiplied gate–drain capacitance of M_6. Since C_{gd6} connects the input and output of the inverting second stage, with gain $-A_{v2}$, it appears at node 1 multiplied by $(1 + A_{v2})$, making it the largest contribution to C_1.

$$C_2 \approx C_{\text{bd6}} + C_{\text{bd7}} + C_{\text{L}} \tag{5}$$

The drain–bulk capacitances of M2 and M4 are read from the operating point. The gate–source and gate–drain capacitances of M6 are estimated by hand.

$$C_{\text{gs6}} \approx \tfrac{2}{3} W_6 L_6 C_{ox} + W_6 L_{ov} C_{ox} \tag{6}$$

$$C_{\text{gd6}} \approx W_6 L_{\text{ov}} C_{\text{ox}} = W_6 C_{\text{ov}} \tag{7}$$

where $C_{ox} = \varepsilon_{ox}/t_{ox} \approx 4.32$ fF/µm^2 ($t_{ox} = 8$ nm) and $L_{ov}C_{ox} = C_{GSO} = 0.1$ fF/µm, both from the GF180MCU model card. For $W_6/L_6 = 13.44/0.56$ µm,

$$\begin{aligned}
C_{\text{gs6}} &\approx 21.7\,\text{fF} + 1.3\,\text{fF} = 23.0\,\text{fF} \\
C_{\text{gd6}} &\approx W_6 C_{\text{ov}} = (13.44\,\mu\text{m})(0.1\,\text{fF}/\mu\text{m}) \approx 1.34\,\text{fF} \\
C_1 &\approx C_{\text{bd2}} + C_{\text{bd4}} + C_{\text{gs6}} + (1 + A_{v2})\, C_{\text{gd6}} \\
&= 5.19\,\text{fF} + 1.71\,\text{fF} + 23.0\,\text{fF} + (121.5)(1.34\,\text{fF}) \approx 193\,\text{fF} \\
C_2 &\approx C_{\text{bd6}} + C_{\text{bd7}} + C_{\text{L}} = 11.5\,\text{fF} + 43.2\,\text{fF} + 1\,\text{pF} = 1.0547\,\text{pF}
\end{aligned}$$

Determining the poles of the system, we have that there are two major poles, one at the output of each stage.

$$\begin{aligned}
f_{\text{p1}} &= \frac{1}{2\pi R_1 C_1} = \frac{1}{2\pi(1.18\,\text{M}\Omega)(193\,\text{fF})} = 699\,\text{kHz} \\
f_{\text{p2}} &= \frac{1}{2\pi R_2 C_2} = \frac{1}{2\pi(178\,\text{k}\Omega)(1.0547\,\text{pF})} = 846.6\,\text{kHz}
\end{aligned}$$

The two poles are close together (699 kHz and 847 kHz), so neither is clearly dominant and both contribute nearly 90° of phase at f_{t}.

Since $f_{\text{t}} = 201$ MHz, we have $f_{\text{p1}}, f_{\text{p2}} < f_{\text{t}}$. Determining the phase margin, we have

$$\text{PM} = 180° - \tan^{-1}\left(\frac{f_{\text{t}}}{f_{\text{p1}}}\right) - \tan^{-1}\left(\frac{f_{\text{t}}}{f_{\text{p2}}}\right) = 180° - 89.8° - 89.76° = 0.44°$$

This differs from the $-8.2°$ phase margin from the simulation. Since this two pole system can at most add 90° phase lag for each pole, the minimum phase margin would be 0°. Therefore, there are poles/zeros at other nodes, such as the current-mirror node at the gates of M_3 and M_4, that contribute the extra phase lag to result in a phase margin of $-8.2°$.

6.2 AC Response With Compensation

With the addition of C_{C}, from the transfer function, we have that [1]:

$$\omega_{\mathrm{z}} = \frac{g_{\mathrm{m6}}}{C_{\mathrm{C}}}$$
$$\omega_{\mathrm{p1}} \approx \frac{1}{R_1 C_{\mathrm{C}}\, g_{\mathrm{m6}} R_2}$$
$$\omega_{\mathrm{p2}} \approx \frac{g_{\mathrm{m6}}}{C_2}$$

To have stability in a unity-gain configuration, we must have a uniform gain roll off at a rate of -20 dB/decade to 0 dB. Therefore, we must have that

$$\omega_{\mathrm{t}} = (g_{\mathrm{m1}} R_1\, g_{\mathrm{m6}} R_2)\, \omega_{\mathrm{p1}} = (g_{\mathrm{m1}} R_1\, g_{\mathrm{m6}} R_2) \left(\frac{1}{R_1 C_{\mathrm{C}}\, g_{\mathrm{m6}} R_2} \right) = \frac{g_{\mathrm{m1}}}{C_{\mathrm{C}}}$$

We must have that $\omega_{\mathrm{t}} < \omega_{\mathrm{z}}$ and $\omega_{\mathrm{t}} < \omega_{\mathrm{p2}}$ [1]. Thus,

$$\omega_{\mathrm{t}} < \omega_{\mathrm{z}} \;\Rightarrow\; \frac{g_{\mathrm{m1}}}{C_{\mathrm{C}}} < \frac{g_{\mathrm{m6}}}{C_{\mathrm{C}}} \;\Rightarrow\; g_{\mathrm{m1}} < g_{\mathrm{m6}} \tag{8}$$

$$\omega_{\mathrm{t}} < \omega_{\mathrm{p2}} \;\Rightarrow\; \frac{g_{\mathrm{m1}}}{C_{\mathrm{C}}} < \frac{g_{\mathrm{m6}}}{C_2} \tag{9}$$

At f_{t}, the total phase lag is,

$$\phi_{\text{phase lag}} = 90^\circ + \tan^{-1}\left(\frac{f_{\mathrm{t}}}{f_{\mathrm{p2}}} \right) + \tan^{-1}\left(\frac{f_{\mathrm{t}}}{f_{\mathrm{z}}} \right)$$

Thus, the phase margin is,

$$\mathrm{PM} = 180^\circ - \phi_{\text{phase lag}} = 90^\circ - \tan^{-1}\left(\frac{g_{\mathrm{m1}}}{C_{\mathrm{C}}} \frac{C_2}{g_{\mathrm{m6}}} \right) - \tan^{-1}\left(\frac{f_{\mathrm{t}}}{f_{\mathrm{z}}} \right)$$

We can simplify determining the phase margin by removing the contribution of f_{z}, for example by placing the zero at infinite frequency. We can implement this by placing a resistor R in series with the compensation capacitor [1]. Finding the zero, we have

$$\omega_{\mathrm{z}} = \frac{1}{C_{\mathrm{C}} \left(\frac{1}{g_{\mathrm{m6}}} - R \right)}$$

Writing the resistor as

$$R = \frac{k}{g_{\mathrm{m6}}} \tag{10}$$

gives $\omega_{\mathrm{z}} = g_{\mathrm{m6}}/[C_{\mathrm{C}}(1-k)]$. With $k = 1$ the zero moves to infinity, and with $k > 1$ it moves into the left half-plane, which adds to the phase margin.

Thus,

$$R = \frac{1}{g_{\mathrm{m6}}} = \frac{1}{6.76 \times 10^{-4}\,\mathrm{S}} = 1479\,\Omega$$

Therefore,

$$\tan^{-1}\left(\frac{f_{\mathrm{t}}}{f_{\mathrm{z}}} \right) = 0$$

We then have

$$\mathrm{PM} = 90^\circ - \tan^{-1}\left(\frac{f_{\mathrm{t}}}{f_{\mathrm{p2}}} \right) = 90^\circ - \tan^{-1}\left(\frac{g_{\mathrm{m1}}}{C_{\mathrm{C}}} \frac{C_2}{g_{\mathrm{m6}}} \right) \tag{11}$$

We target a phase margin of 60°. Thus,

$$\mathrm{PM} = 90^\circ - \tan^{-1}\left(\frac{g_{\mathrm{m1}}}{C_{\mathrm{C}}}\frac{C_2}{g_{\mathrm{m6}}}\right)$$

$$\Rightarrow\ C_{\mathrm{C}} = \frac{g_{\mathrm{m1}}}{\tan(90^\circ - \mathrm{PM})}\frac{C_2}{g_{\mathrm{m6}}} = g_{\mathrm{m1}}\frac{C_2}{g_{\mathrm{m6}}}\tan(\mathrm{PM}) = 233.8\,\mathrm{fF}$$

Thus,

$$f_{\mathrm{t}} = \frac{1}{2\pi}\frac{g_{\mathrm{m1}}}{C_{\mathrm{C}}} = 58.9\,\mathrm{MHz}$$

Confirming the conditions from Eqs. (8) and (9), we have

$$g_{\mathrm{m1}} = 0.0865\,\mathrm{mS} < g_{\mathrm{m6}} = 0.676\,\mathrm{mS}$$

$$\frac{g_{\mathrm{m1}}}{C_{\mathrm{C}}} = \frac{0.0865\times10^{-3}\,\mathrm{S}}{233.8\times10^{-15}\,\mathrm{F}} = 369.97\times10^{6}\,\mathrm{rad/s} < \frac{g_{\mathrm{m6}}}{C_2} = \frac{0.676\times10^{-3}\,\mathrm{S}}{1.0547\times10^{-12}\,\mathrm{F}} = 640.9\times10^{6}\,\mathrm{rad/s}$$

Figure 5 shows the schematic with the compensation capacitor and resistor added.

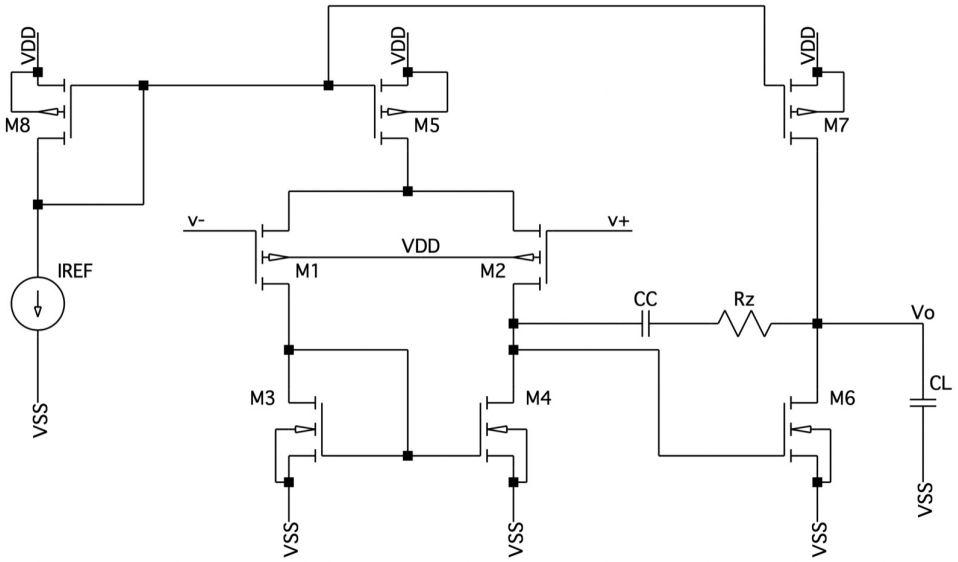


Figure 5: Schematics of Two-Stage Op Amp with Compensation Capacitor and Resistor

Performing the AC analysis as before, with the inductor L_{FB} and capacitor C_{FB}, we have the following frequency response in Fig. 6:

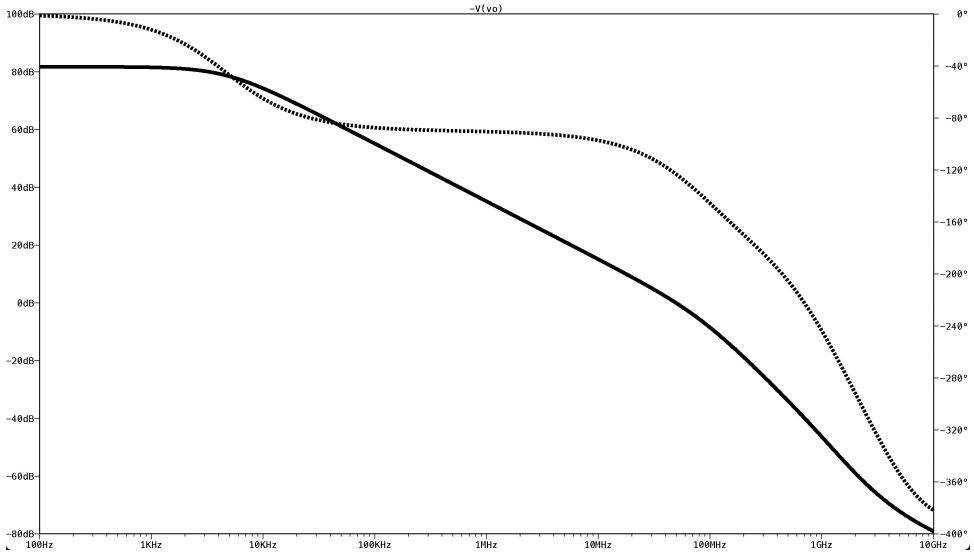


Figure 6: Frequency Response of Two-Stage Op Amp with $C_C = 233.8\,\text{fF}$ and Ideal Resistor

The DC gain is 81.74 dB. The simulated f_t is 49.6 MHz, less than the predicted 58.9 MHz because of parasitic capacitance and the second pole. The phase at f_t is approximately $-123.07°$. Thus, the phase margin is now

$$\text{PM} = 180° - 123.07° = 56.93°.$$

This is a significant increase in the phase margin from $-8.2°$ and close to the target phase margin of 60°. The $-3\,\text{dB}$ bandwidth is 4.69 kHz. From this we see that the phase margin has increased and the phase from the right-half-plane zero has been removed. This is at the cost of the open loop bandwidth decreasing significantly. However, this is acceptable as the main use of differential amplifiers is in a negative feedback configuration, which increases the bandwidth, while decreasing the gain.

6.3 Implementing Resistance Using NMOS In Triode Region

From Eq. (10), we have that

$$R_\text{Z} = \frac{k}{g_\text{m6}}$$

where $k = 1$ for moving the zero to infinity.

For M_6 in saturation we have

$$g_\text{m6} = \mu_n C_\text{ox} \left(\frac{W}{L}\right)_6 V_\text{OV6}$$

For an NMOS, M_Z, in the triode region we have

$$I_\text{DZ} = \mu_n C_\text{ox} \left(\frac{W}{L}\right)_Z \left[V_\text{OVZ} V_\text{DSZ} - \frac{V_\text{DSZ}^2}{2}\right] \approx \mu_n C_\text{ox} \left(\frac{W}{L}\right)_Z V_\text{OVZ} V_\text{DSZ}$$

for $V_\text{DSZ} \approx 0$. Thus, the conductance is

$$\begin{aligned} g_\text{dsZ} &= \left.\frac{\partial I_\text{DZ}}{\partial V_\text{DSZ}}\right|_{V_\text{DSZ}\approx 0} = \mu_n C_\text{ox} \left(\frac{W}{L}\right)_Z V_\text{OVZ} \\ &= \frac{1}{R_\text{Z}} \end{aligned}$$

Multiplying g_{m6} by $R_{\mathrm{Z}} = 1/g_{\mathrm{dsZ}}$, we have

$$g_{\mathrm{m6}} R_{\mathrm{Z}} = \frac{W_6}{W_{\mathrm{Z}}} \frac{V_{\mathrm{OV6}}}{V_{\mathrm{OVZ}}} = k$$

If we have $V_{\mathrm{OV6}} = V_{\mathrm{OVZ}}$, then only the widths determine the ratio.

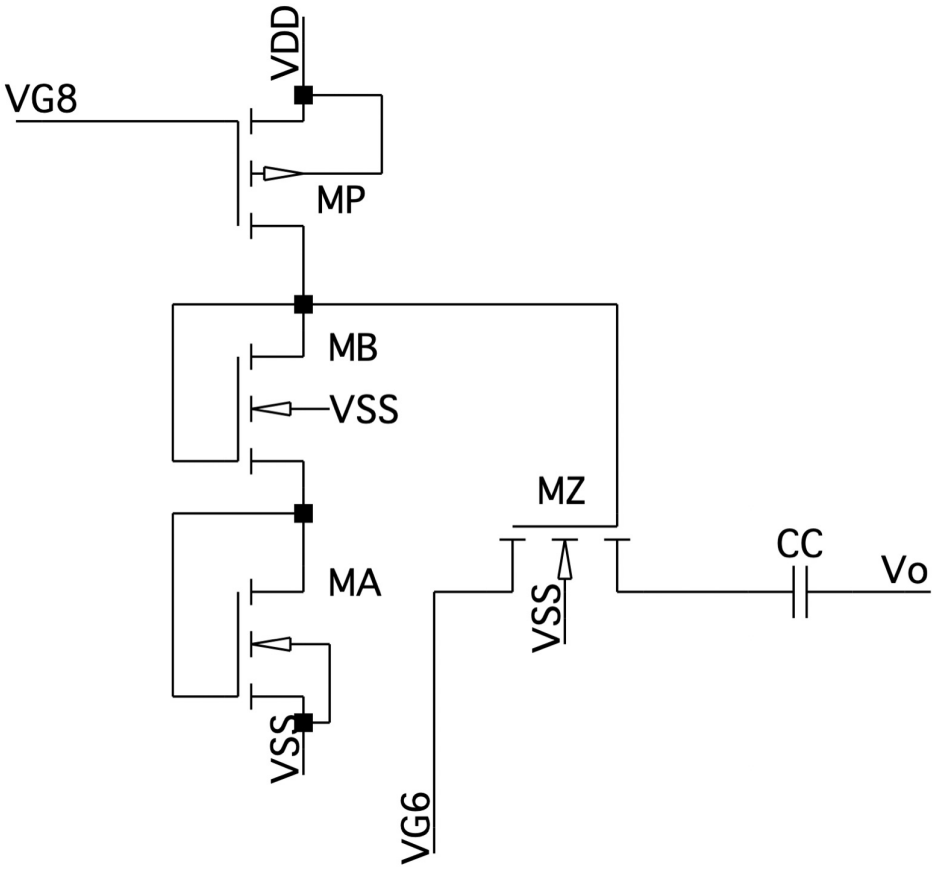


Figure 7: Implementing Resistance with NMOS in Triode (after [2])

Figure 7 shows a schematic of how the resistance can be implemented using an NMOS, M_{Z}, in the triode region [2]. It shows two stacked diode-connected NMOS transistors, M_{A}, M_{B}, connected to a PMOS M_{P}. The source of M_{Z} is connected to the gate of M_6. The gate of M_{Z} is connected to the drain and gate of M_{B}. Thus,

$$V_{\mathrm{GZ}} = V_{\mathrm{GSA}} + V_{\mathrm{GSB}}$$
$$V_{\mathrm{SZ}} = V_{\mathrm{GS6}}$$

Thus,

$$V_{\mathrm{GSZ}} = V_{\mathrm{GSA}} + V_{\mathrm{GSB}} - V_{\mathrm{GS6}}$$

We size M_{A} and M_{B} at the same current density as M_6. Since M_{A} and M_6 have their sources connected to $-V_{\mathrm{SS}}$, then $V_{SB} = 0$ for both and they have the same threshold and, at equal current density, the same overdrive, giving

$$V_{\mathrm{GSA}} = V_{\mathrm{GS6}} \;\Rightarrow\; V_{\mathrm{GSZ}} = V_{\mathrm{GSA}} + V_{\mathrm{GSB}} - V_{\mathrm{GS6}} = V_{\mathrm{GSB}}$$

Both M_{B} and M_{Z} have their sources one V_{GS} above $-V_{\mathrm{SS}}$ and will be impacted by the body effect. Thus,

$$V_{\mathrm{thB}} \approx V_{\mathrm{thZ}}$$

Determining V_{OVZ}, we have

$$\begin{aligned} V_{\mathrm{OVZ}} &= (V_{\mathrm{GSA}} + V_{\mathrm{GSB}} - V_{\mathrm{GS6}}) - V_{\mathrm{thZ}} \\ &= (V_{\mathrm{thA}} - V_{\mathrm{th6}}) + (V_{\mathrm{thB}} - V_{\mathrm{thZ}}) + V_{\mathrm{OVA}} + V_{\mathrm{OVB}} - V_{\mathrm{OV6}}. \end{aligned}$$

M_{A} and M_6 have no body effect, so $V_{\mathrm{thA}} = V_{\mathrm{th6}}$; M_{B} and M_{Z} have the same body effect, so $V_{\mathrm{thB}} \approx V_{\mathrm{thZ}}$; and equal current density gives $V_{\mathrm{OVA}} = V_{\mathrm{OV6}}$. Thus

$$V_{\mathrm{OVZ}} = V_{\mathrm{OVB}} \approx V_{\mathrm{OV6}}$$

and

$$k = \frac{W_6}{W_{\mathrm{Z}}} \frac{V_{\mathrm{OV6}}}{V_{\mathrm{OVZ}}} \approx \frac{W_6}{W_{\mathrm{Z}}}.$$

Sizing M_{P} Since we require the same current density as PMOS M_8, we have

$$W_{\mathrm{P}} = W_8 \frac{I_{\mathrm{P}}}{I_{\mathrm{REF}}}$$

We only require a fraction of I_{REF}. Thus, we choose $I_{\mathrm{P}} = 8\,\mu\mathrm{A}$. Thus,

$$W_{\mathrm{P}} = (13.7\,\mu\mathrm{m}) \frac{8\,\mu\mathrm{A}}{20\,\mu\mathrm{A}} = 5.48\,\mu\mathrm{m}$$

Sizing M_{A} and M_{B} Since we require the same current density as NMOS M_6, we have

$$W_{\mathrm{A,B}} = W_6 \frac{I_{\mathrm{P}}}{I_6} = (13.44\,\mu\mathrm{m}) \frac{8\,\mu\mathrm{A}}{80\,\mu\mathrm{A}} = 1.344\,\mu\mathrm{m}$$

Sizing M_{Z} Since $V_{\mathrm{OV6}} = V_{\mathrm{OVZ}} = V_{\mathrm{OVA,B}}$,

$$k = g_{\mathrm{m6}} R_{\mathrm{Z}} = \frac{W_6}{W_{\mathrm{Z}}} \frac{V_{\mathrm{OV6}}}{V_{\mathrm{OVZ}}} = \frac{W_6}{W_{\mathrm{Z}}} \;\Rightarrow\; W_{\mathrm{Z}} = \frac{W_6}{k}$$

We select $k = 1.2$ instead of 1 to allow for some margin. Thus,

$$W_{\mathrm{Z}} = \frac{W_6}{1.2} = 11.2\,\mu\mathrm{m}$$

Performing a DC operating point of the circuit with the transistors M_P, M_B, M_A, M_Z, we have the following in Table 3:

Table 3: DC operating point parameters of the bias network

Transistor	M_{P}	M_{B}	M_{A}	M_{Z}	M_6
I_{D}	8.11 μA	8.11 μA	8.11 μA	1.78 pA	81.9 μA
$\lvert V_{\mathrm{GS}}\rvert$ (V)	0.98	1.161	0.895	1.167	0.889
$\lvert V_{\mathrm{DS}}\rvert$	1.24 V	1.161 V	0.895 V	1.50 nV	1.63 V
$\lvert V_{\mathrm{BS}}\rvert$ (V)	0	0.895	0	0.889	0
$\lvert V_{\mathrm{th}}\rvert$ (V)	0.788	0.961	0.696	0.969	0.708
$\lvert V_{\mathrm{dsat}}\rvert$ (V)	0.195	0.227	0.215	0.225	0.203
g_{m}	70.2 μS	65.2 μS	64.6 μS	4.15 pS	676 μS
g_{mb}	31.8 μS	16.4 μS	23.7 μS	1.07 pS	246 μS
g_{ds}	0.291 μS	0.399 μS	0.487 μS	591 μS	3.23 μS
C_{bd} (fF)	4.64	1.20	1.39	10.7	11.5
C_{bs} (fF)	5.87	1.39	1.73	10.7	16.0

Thus,

$$\frac{1}{g_{\mathrm{m6}}} = 1479\,\Omega$$

$$R_{\mathrm{Z}} = \frac{1}{g_{\mathrm{dsZ}}} = 1692\,\Omega$$

$$k = g_{\mathrm{m6}} R_{\mathrm{Z}} = 1.144$$

This is close to the desired value of 1.2. To bring k closer to 1.2, we have

$$W_{\mathrm{Z,new}} = 11.2\,\mu\mathrm{m} \times \frac{1.144}{1.2} \approx 10.7\,\mu\mathrm{m}$$

Performing a DC operating point simulation with this new sizing, we have the following in Table 4:

Table 4: DC operating point parameters of the bias network with resized M_{Z}

Transistor	M_{P}	M_{B}	M_{A}	M_{Z}	M_6
I_{D}	8.11 μA	8.11 μA	8.11 μA	1.78 pA	81.9 μA
$\lvert V_{\mathrm{GS}}\rvert$ (V)	0.98	1.161	0.895	1.167	0.889
$\lvert V_{\mathrm{DS}}\rvert$	1.24 V	1.161 V	0.895 V	1.57 nV	1.63 V
$\lvert V_{\mathrm{BS}}\rvert$ (V)	0	0.895	0	0.889	0
$\lvert V_{\mathrm{th}}\rvert$ (V)	0.788	0.961	0.696	0.969	0.708
$\lvert V_{\mathrm{dsat}}\rvert$ (V)	0.195	0.227	0.215	0.225	0.203
g_{m}	70.2 μS	65.2 μS	64.6 μS	4.15 pS	676 μS
g_{mb}	31.8 μS	16.4 μS	23.7 μS	1.07 pS	246 μS
g_{ds}	0.291 μS	0.399 μS	0.487 μS	565 μS	3.23 μS
C_{bd} (fF)	4.64	1.20	1.39	10.2	11.5
C_{bs} (fF)	5.87	1.39	1.73	10.2	16.0

Thus,

$$\frac{1}{g_{\mathrm{m6}}} = 1479\,\Omega$$

$$R_{\mathrm{Z}} = \frac{1}{g_{\mathrm{dsZ}}} = 1770\,\Omega$$

$$k = g_{\mathrm{m6}} R_{\mathrm{Z}} = 1.20$$

Thus, we achieve the desired value. Performing the AC analysis, we have the following frequency response in Fig. 8:

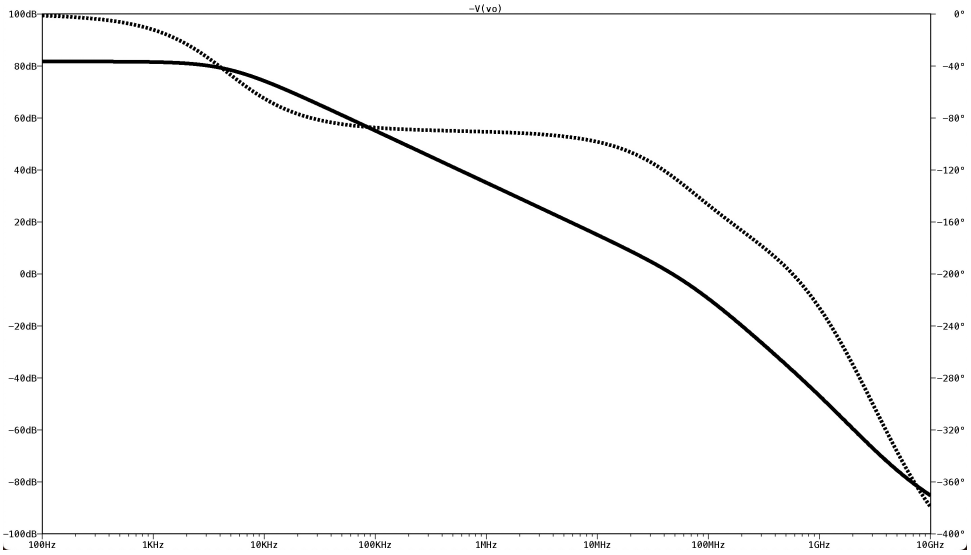


Figure 8: Frequency Response of Two-Stage Op Amp with Compensation Capacitor $C_C = 233.8\,\text{fF}$ and NMOS In Triode

The DC gain is 81.74 dB. We also have $f_\text{t} = 47.5\,\text{MHz}$, and the phase at f_t is approximately $-124.96°$. Thus, the phase margin is now

$$\text{PM} = 180° - 124.96° = 55.04°.$$

This is still close to the target phase margin of 60°. The $-3\,\text{dB}$ bandwidth is 4.68 kHz.

7 Time-Domain Behaviour and Slew Rate

The step responses use the op-amp of Section B.2 connected as a unity-gain buffer (the gate of M_1 tied to the output), driven through 1 kΩ by `PULSE(0.8 1.8 0.1u 100p 100p 200n)`.

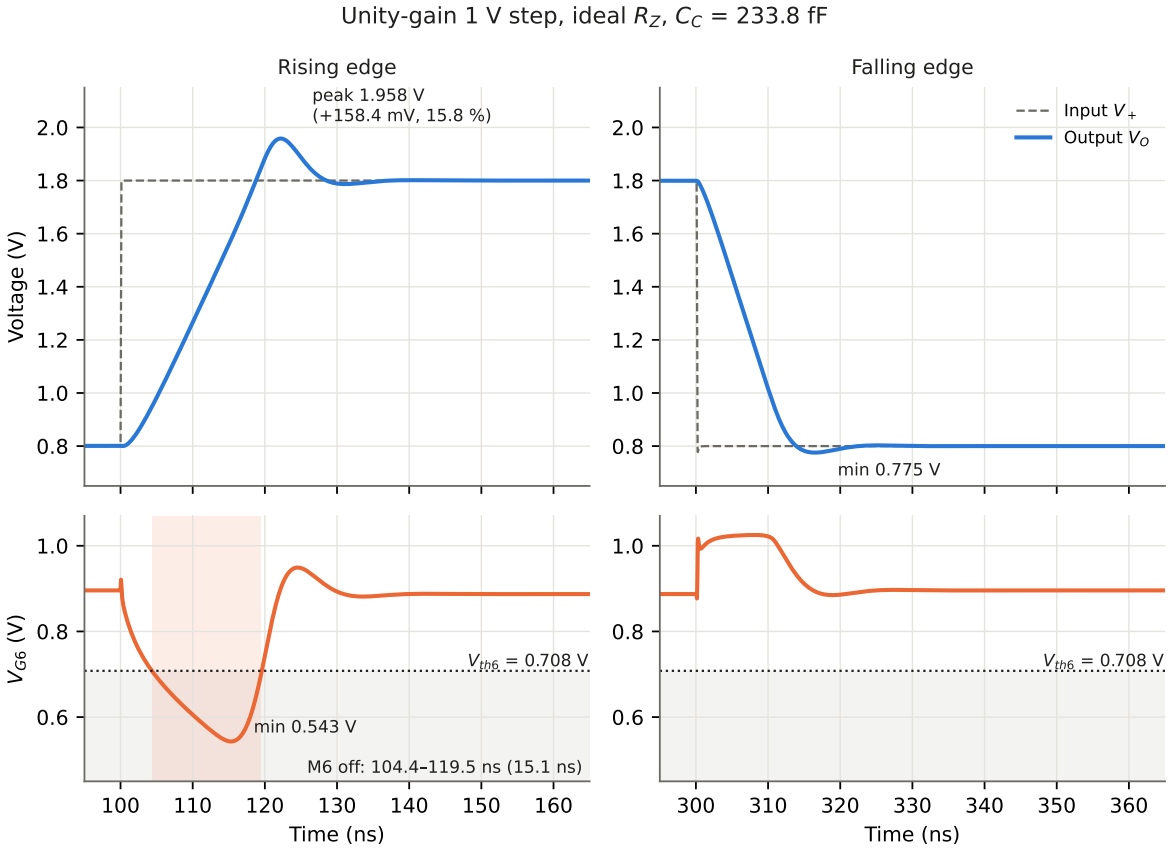


Figure 9: Large step response with $C_C = 233.8\,\text{fF}$ (ideal R_Z). Top: output V_O and input V_+. Bottom: gate voltage of M_6. V_{G6} falls to 0.543 V, below $V_{tn6} = 0.708\,\text{V}$, so M_6 is off from 104.4 to 119.5 ns and the output overshoots by 158 mV (15.8%).

We see a large overshoot in the response of 0.158 V, which is 15.8%. This large overshoot can be attributed to the fact that V_{G6} is pulled down and turns M_6 off.

For M_6 to stay on, we need $V_{GS6} = V_{G6} > V_{tn6} = 0.708\,\text{V}$. In Fig. 9, we see that V_{G6} drops below 0.708 V from 104.4 ns to 119.5 ns. We have that the slew rate is

$$\text{SR} = \frac{I}{C_C} = \frac{19.5\,\mu\text{A}}{233.8\,\text{fF}} = 83\,\text{V}/\mu\text{s}$$

This assumes that I_7 is sufficient to provide current I to C_C and $\text{SR} \cdot C_2$ to C_2 . This means that the first stage limits the slew rate [1]. Thus,

$$I_7 \geq I + \text{SR} \cdot C_2 = I + \frac{I}{C_C} \cdot C_2 = I\left(1 + \frac{C_2}{C_C}\right) \tag{12}$$

From Eq. (12), we have

$$\frac{I_7 - I}{C_2} \geq \frac{I}{C_C} \tag{13}$$

From Eq. (13), the output slew rate, $\frac{I_7 - I}{C_2}$, must be greater than or equal to the first stage slew rate, $\frac{I}{C_C}$.

From Eq. (12), we have

$$I\left(1+\frac{C_2}{C_\mathrm{C}}\right) = (19.5\,\mu\mathrm{A})\left(1+\frac{1.0547\,\mathrm{pF}}{233.8\,\mathrm{fF}}\right) = 107.5\,\mu\mathrm{A}$$

But $I_7 = 81.9\,\mu\mathrm{A} < 107.5\,\mu\mathrm{A}$. Thus, the condition is violated and the output limits the slew rate to

$$\frac{I_7 - I}{C_2} = \frac{81.9\,\mu\mathrm{A} - 19.5\,\mu\mathrm{A}}{1.0547\,\mathrm{pF}} = 59\,\mathrm{V}/\mu\mathrm{s}$$

Rearranging Eq. (13), we have

$$\frac{I}{I_7 - I}\,C_2 = 330\,\mathrm{fF} \le C_\mathrm{C}$$

From Section 6.2, we have that

$$C_\mathrm{C} = g_{\mathrm{m}1}\,\frac{C_2}{g_{\mathrm{m}6}}\,\tan(\mathrm{PM}) \;\Rightarrow\; \frac{C_2}{C_\mathrm{C}} = \frac{g_{\mathrm{m}6}}{g_{\mathrm{m}1}\tan(\mathrm{PM})}$$

where PM is the required phase margin. Thus,

$$I_7 \ge I\left(1+\frac{C_2}{C_\mathrm{C}}\right) = I\left(1+\frac{g_{\mathrm{m}6}}{g_{\mathrm{m}1}\tan(\mathrm{PM})}\right)$$

Therefore,

$$\tan(\mathrm{PM}) \ge \frac{g_{\mathrm{m}6}}{g_{\mathrm{m}1}}\,\frac{I}{I_7 - I} = 2.44 \;\Rightarrow\; \mathrm{PM} \ge \arctan(2.44) = 67.7^\circ$$

Thus, to meet the slewing requirements with $I_7 = 81.9\,\mu\mathrm{A}$, a phase margin of 67.7° is needed. This corresponds to $C_\mathrm{C} = 330\,\mathrm{fF}$. We can therefore increase C_C so that the first stage limits slewing, and we choose 350 fF to have some margin. This will lower $\omega_{\mathrm{p}1}$, and ω_t as well.

Performing the step response using the op amp of Section B.2 in a unity gain feedback configuration with C_C=350 fF we have the following in Fig. 10.

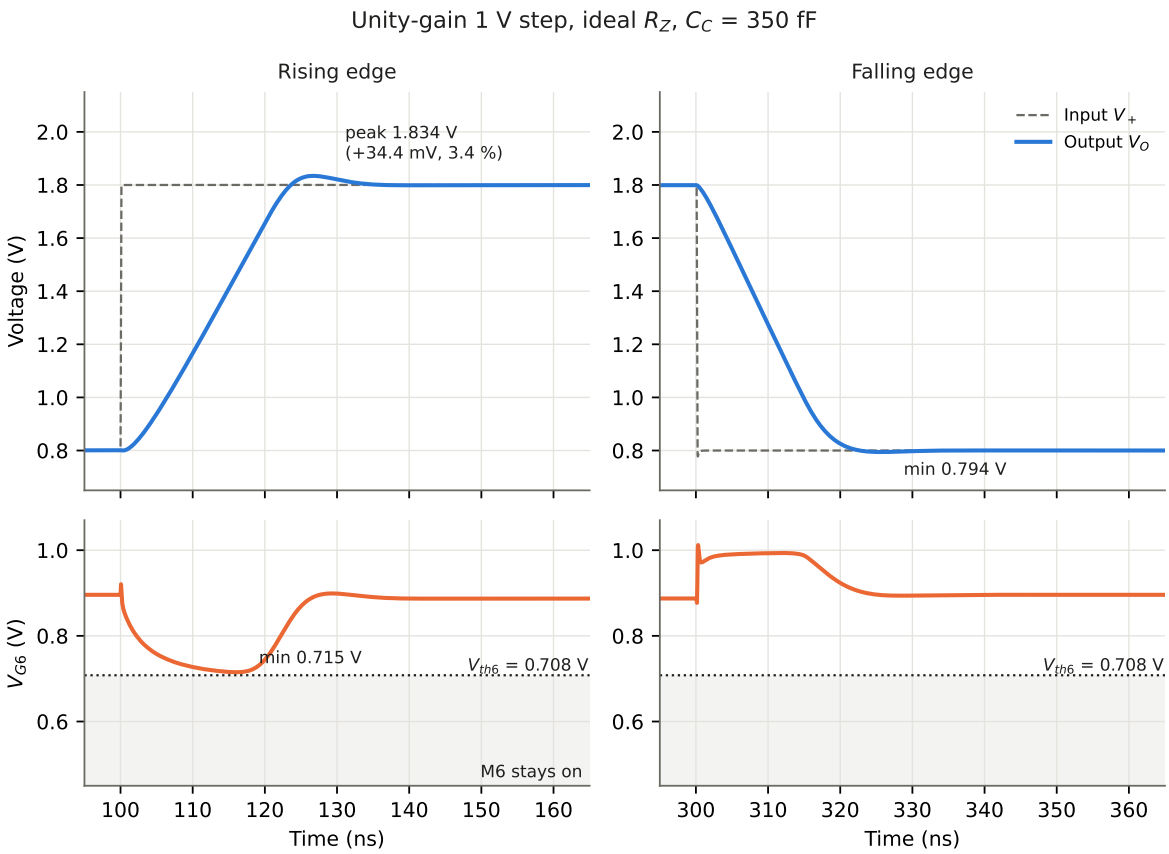


Figure 10: Large step response with $C_C = 350\,\text{fF}$ (ideal R_Z). V_{G6} reaches a minimum of 0.715 V, just above V_{tn6}, so M_6 stays on and the overshoot is 34.4 mV (3.44%).

We see a very small overshoot in the response of 34.4 mV, which is 3.44%. V_{G6} stays above 0.715 V, just above $V_{tn6} = 0.708\,\text{V}$, so M_6 never turns off. Table 5 summarizes the findings for slew rate.

Table 5: Slew Rate and Transient Performance for Two Values of C_C

	$C_C = 233.8\,\text{fF}$	$C_C = 350\,\text{fF}$
Measured Rising Slew Rate (V/μs)	57.6	47.5
Measured Falling Slew Rate (V/μs)	82.5	54.8
First-Stage Limit I/C_C (V/μs)	83	56
Second-Stage Limit $(I_7 - I)/C_2$ (V/μs)	59	59
Rise Time, 10% to 90% (ns)	13.9	16.8
Fall Time, 90% to 10% (ns)	9.7	14.6

For $C_C = 233.8\,\text{fF}$, we see that the rising slew rate is limited by the output and the falling slew rate is limited by the first stage. For $C_C = 350\,\text{fF}$, both the rising and falling slew rates are limited by the first stage.

The large overshoot is reduced, but the rise and fall times increased. Performing AC analysis with C_C=350 fF, we have the response in Fig. 11, using the inductor L_{FB} and capacitor C_{FB}:

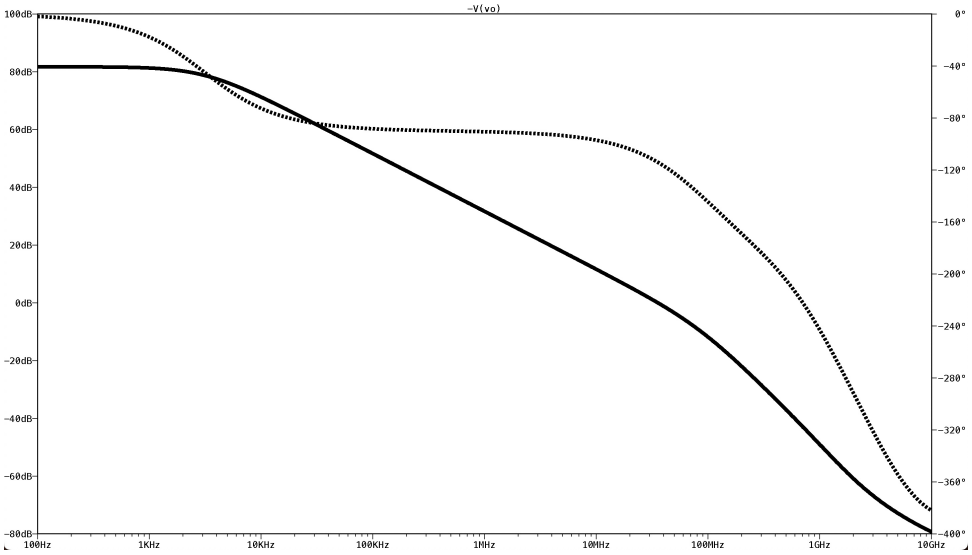


Figure 11: Frequency Response of Two-Stage Op Amp with 350 fF Compensation Capacitor and R_Z

Table 6 summarizes the findings for the AC response using the ideal resistor R_{Z}.

Table 6: AC Response for Two Values of C_{C} (Ideal R_{Z})

	$C_{\mathrm{C}} = 233.8\,\mathrm{fF}$	$C_{\mathrm{C}} = 350\,\mathrm{fF}$
DC Gain	81.7 dB	81.7 dB
-3 dB Frequency	4.69 kHz	3.14 kHz
f_{t}	49.6 MHz	35.6 MHz
Phase Margin	56.9°	65.9°
f_{180}	268.7 MHz	274.0 MHz
Gain Margin	23.5 dB	27.0 dB

8 Final Design Performance

Figure 12 shows the schematic for the final design of the op amp using the triode transistor to implement the resistor.

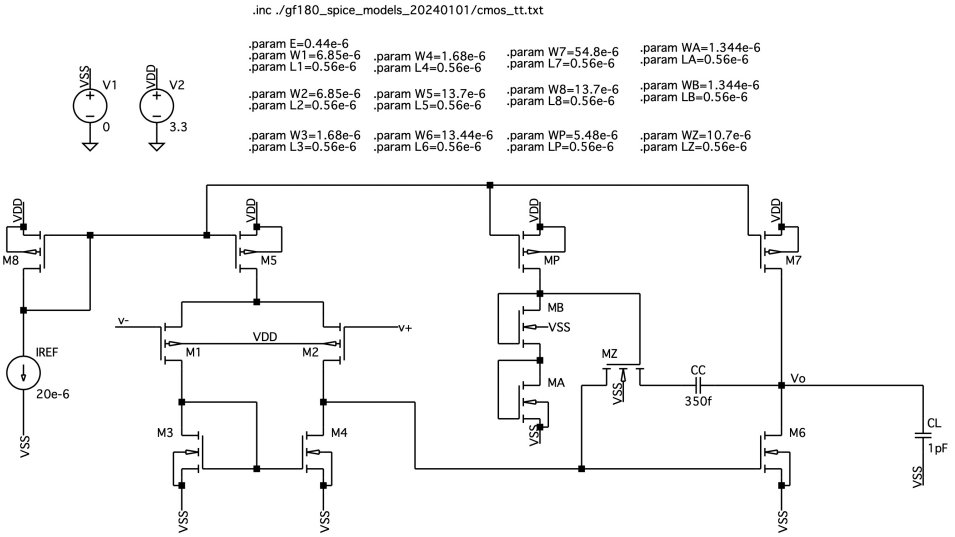


Figure 12: Schematic of Two-Stage Op Amp with Compensation Capacitor and NMOS In Triode

Figure 13 shows the schematic for the final design of the op amp using a resistor.

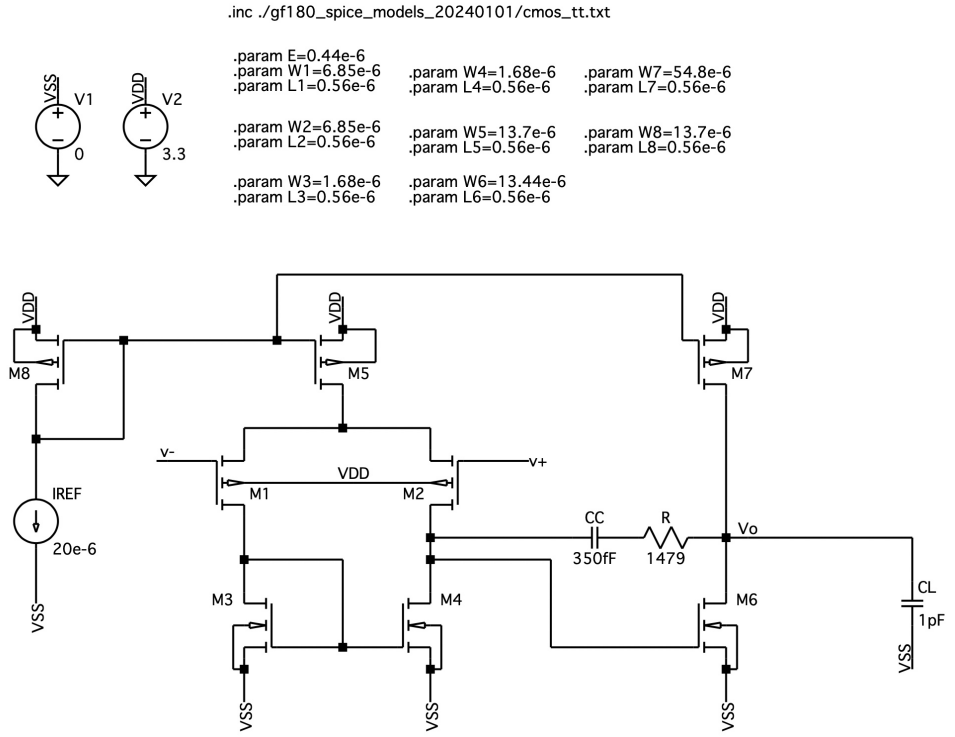


Figure 13: Schematic of Two-Stage Op Amp with Compensation Capacitor and Resistor

8.1 Frequency response

Performing the AC analysis using the inductor and capacitor to set the DC operating point again, we have the following frequency response in Fig. 14 for the triode implementation. The complete netlist is given in Section B.1.

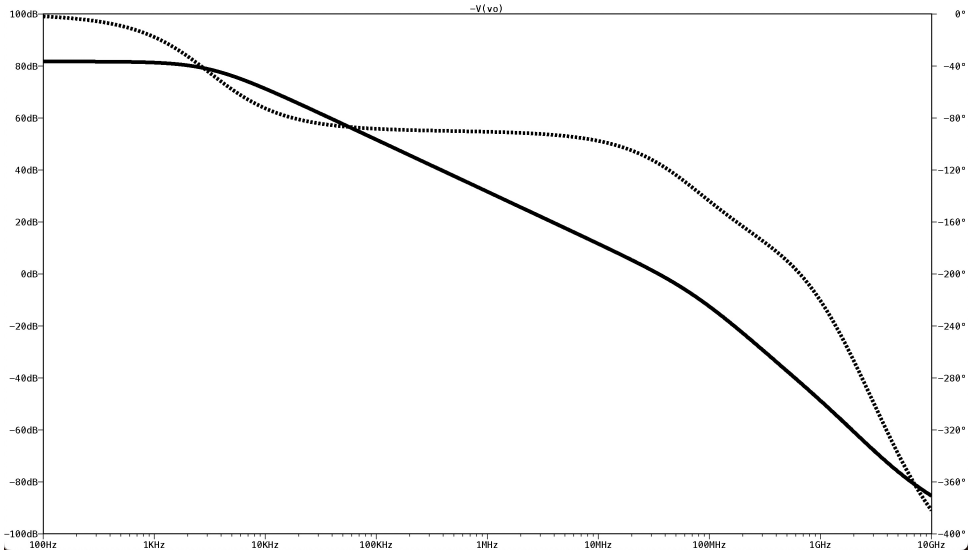


Figure 14: AC Response for the Triode Implementation

Figure 15 shows the AC response for the resistor implementation

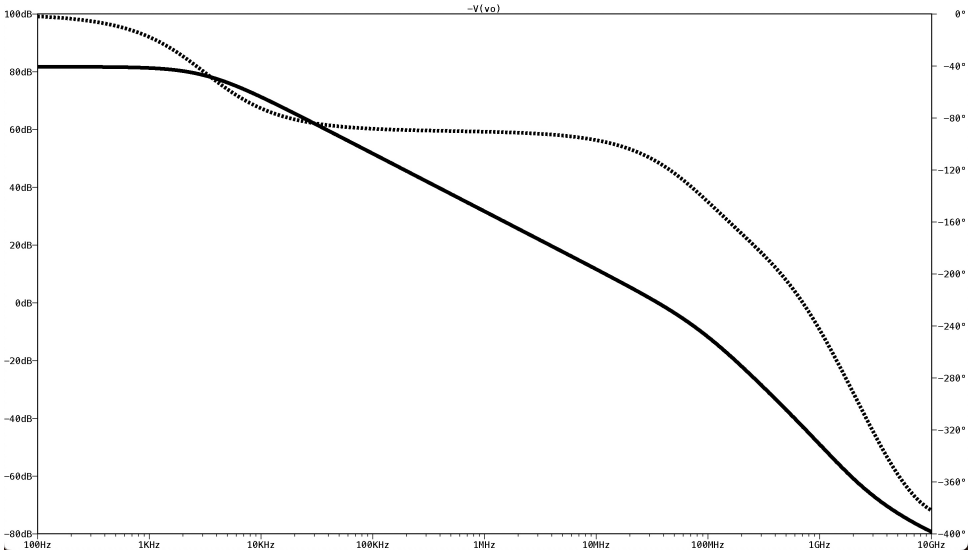


Figure 15: AC Response for the Ideal Resistor Implementation

Table 7 summarizes the findings.

Table 7: AC Response with Ideal Resistor and with M_Z ($C_C = 350\,\text{fF}$)

	Resistor, $R = 1479\,\Omega$	M_Z
$R_Z\, g_{m6}$	1	1.2
DC Gain	81.7 dB	81.7 dB
-3 dB Frequency	3.14 kHz	3.14 kHz
f_t	35.6 MHz	34.9 MHz
Phase Margin	65.9°	64.9°
$-180°$ Frequency	274.0 MHz	368.5 MHz
Gain Margin	27.0 dB	32.4 dB

8.2 Slew Rate

Figure 16 shows the large-step response for the triode implementation

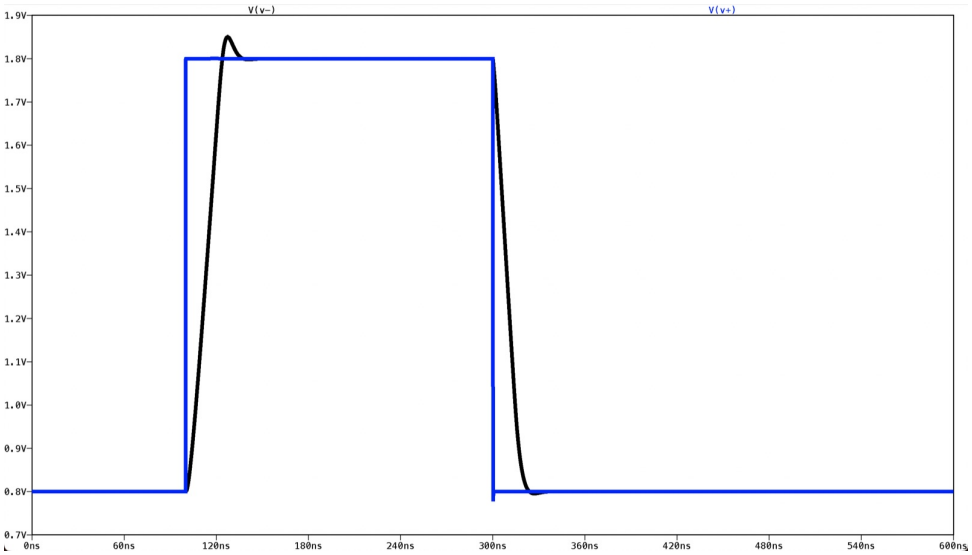


Figure 16: Large Step Response for the Triode implementation

Figure 17 shows the large-step response for the resistor implementation

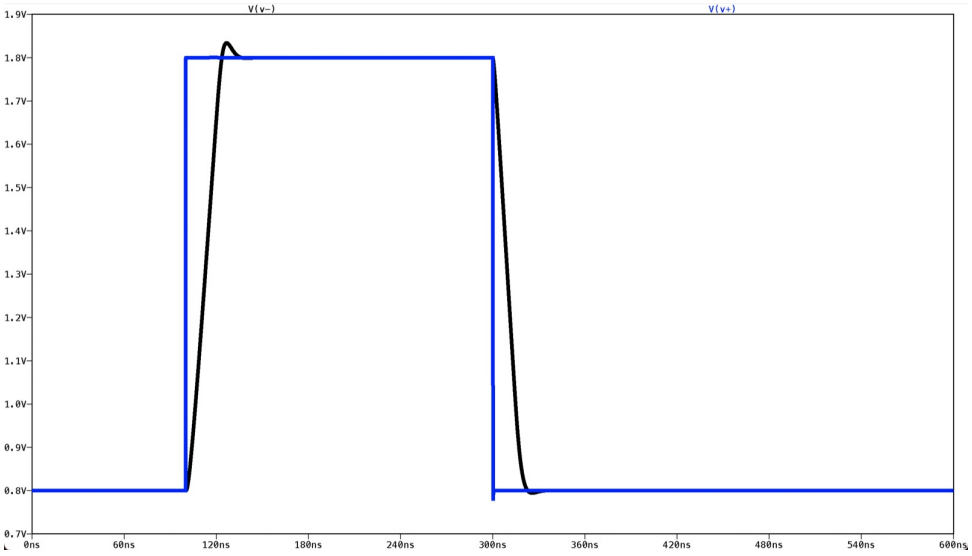


Figure 17: Large Step Response for the Resistor implementation

Table 8 summarizes the findings.

Table 8: Large-Step Response with Ideal Resistor and with M_{Z} ($C_{\mathrm{C}} = 350\,\mathrm{fF}$)

	Resistor, $R = 1479\,\Omega$	M_{Z}
Rising Overshoot	34.4 mV (3.44%)	50.5 mV (5.05%)
Falling Undershoot	5.7 mV (0.57%)	4.6 mV (0.46%)
Rise Time, 10% to 90%	16.8 ns	17.2 ns
Fall Time, 90% to 10%	14.6 ns	14.7 ns
Rising Slew Rate (V/μs)	47.5	46.5
Falling Slew Rate (V/μs)	54.8	54.4
V_{G6} Range	0.715 V to 1.012 V	0.719 V to 1.019 V
I_6 Range	10.7 μA to 187.6 μA	11.4 μA to 194.1 μA
V_{GZ} Range	–	2.032 V to 2.096 V

8.3 CMRR

Figure 18 shows the differential gain, the common-mode gain and the resulting CMRR of the M_Z design. The low-frequency CMRR is 83.1 dB, with a common-mode gain of -1.4 dB. The CMRR stays within 0.1 dB of its low-frequency value up to about 0.6 MHz and the -3 dB point is at 3.7 MHz. A_{cm} reaches its lowest point, -61.8 dB, at about 9 MHz, then rises to -54.6 dB at about 110 MHz, while A_{dm} continues to fall. The CMRR falls below 0 dB only above about 2 GHz, far beyond $f_t = 34.9$ MHz. Because the simulation assumes perfectly matched devices, this value is an upper bound. Device mismatch would reduce it.

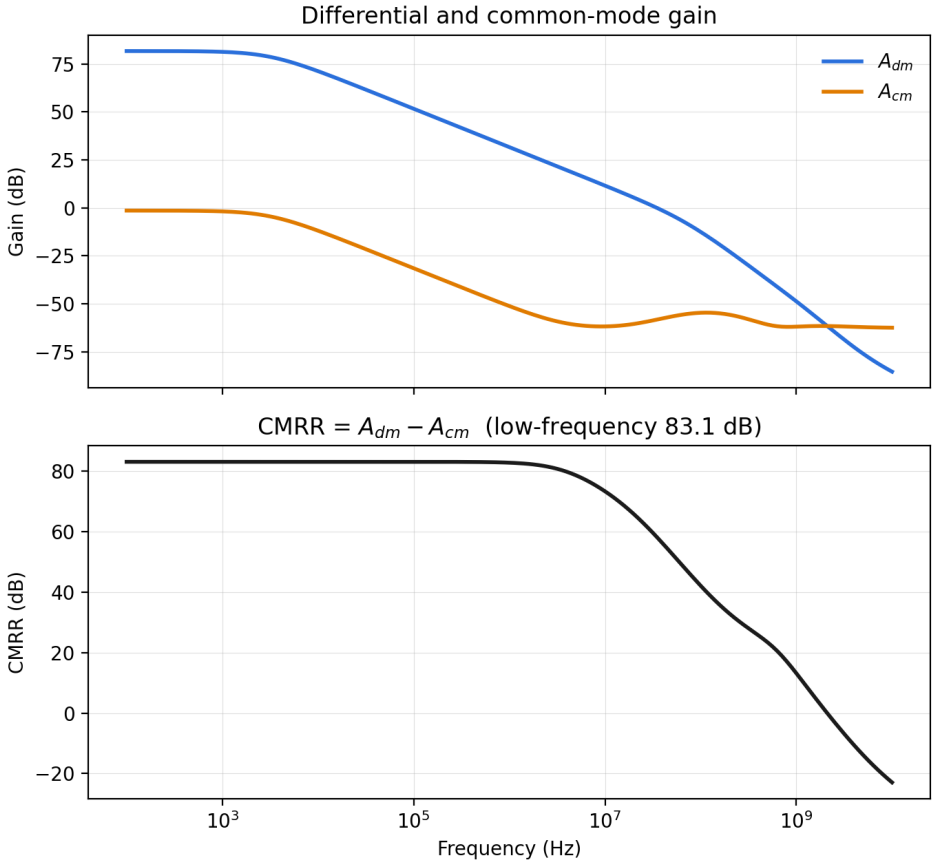


Figure 18: Differential gain A_{dm}, common-mode gain A_{cm} (top) and CMRR $= A_{dm} - A_{cm}$ (bottom), M_Z design, $C_C = 350$ fF, TT, 27 °C.

8.4 Power

Table 9 shows the power consumption summary for the op amp using the triode implementation.

Table 9: Power Consumption by Current Path, M_Z Design, TT, 27 °C, $V_{DD} = 3.3$ V

Branch	Current Path ($V_{DD} \to V_{SS}$)	Current (μA)	Power (μW)	Share
Bias reference	$M_8 \to I_{\mathrm{REF}}$	20.0	66.0	15.5%
First stage	$M_5 \to M_{1,2} \to M_{3,4}$	19.5	64.2	15.0%
Second stage	$M_7 \to M_6$	81.8	269.9	63.2%
M_Z gate bias	$M_P \to M_B \to M_A$	8.1	26.8	6.3%
Total		**129.4**	**426.9**	100%

Table 10 summarizes the power consumption of both designs.

Table 10: Power Consumption of Both Designs

	Resistor, $R = 1479\,\Omega$	M_Z
Power	0.400 mW	0.427 mW

The total power is 0.427 mW from a 3.3 V supply (Table 9). The second stage draws 81.8 μA, nearly two-thirds of the total, since it must supply current to both C_C and the load during slewing. The first stage and the bias reference each draw about 20 μA. The gate bias branch for M_Z draws 8.1 μA (6.3%). This branch is the only difference in power between the two designs. Without this branch, the supply current is 121.3 μA, giving the 0.400 mW of the resistor design.

9 Robustness over Process and Temperature

9.1 Corners

The corner and temperature results were simulated in ngspice-47 with the GF180MCU model library (`sm141064.ngspice`). We vary the process corner (TT, SS, FF, SF, FS), the temperature, and, for the resistor design, the resistor corner, as shown in Table 11. All results use the inductor/capacitor bench of Section 6, with $C_C = 350$ fF. The netlists are given in Section B.3. The resistor design uses the PDK poly resistor (`ppolyf_u_1k`, $W = 2\,\mu$m, $L = 2.75\,\mu$m) rather than an ideal resistor, so that its process corners and temperature coefficient are included. The full results are listed in Section A. R_Z is the on-resistance $1/g_{ds}$ of M_Z or the value of the poly resistor, and $k = g_{m6}R_Z$.

Table 11: Variables Swept in the Corner Analysis

Variable	Values
Process Corner	TT, SS, FF, SF, FS
Temperature	-40, 27, 85, 125 °C
Poly R_Z Corner	typical, `res_ss` (+20% sheet resistance), `res_ff` (-20%)
Poly R_Z at TT, 27 °C	1200, 1478, 1757 Ω

Table 12 shows the phase margin by corner.

Table 12: Phase Margin by Corner, Across All Temperatures (ngspice)

Corner	Poly R_Z, `res_ff` to `res_ss`	Spread	M_Z	Spread
TT	63.0° to 70.5°	7.4°	65.3° to 65.8°	0.5°
SS	62.7° to 69.9°	7.2°	65.4° to 65.7°	0.3°
FF	63.3° to 70.9°	7.6°	65.1° to 65.8°	0.7°
SF	61.6° to 69.2°	7.6°	64.2° to 64.7°	0.5°
FS	64.4° to 71.6°	7.2°	66.3° to 66.8°	0.5°
Overall	61.6° to 71.6°	10.0°	64.2° to 66.8°	2.6°

Figure 19 shows the phase margin against temperature for both designs.

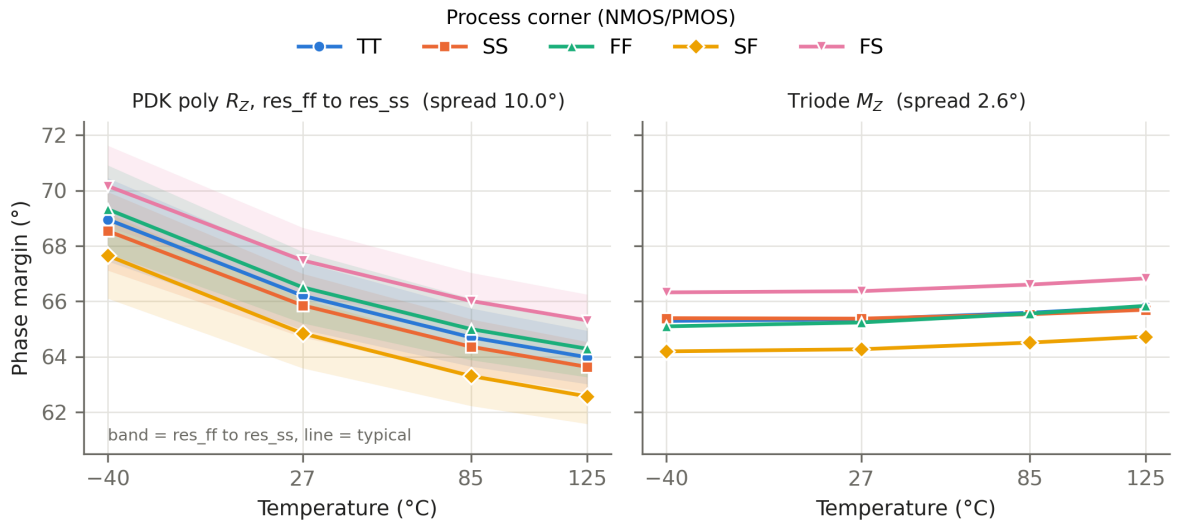


Figure 19: Phase margin versus temperature for the five process corners (ngspice). Left: PDK poly R_Z, typical sheet resistance (line) and `res_ff` to `res_ss` (band). Right: M_Z. With the poly resistor, the phase margin falls by about 5° from −40 to 125 °C at every corner, giving a total spread of 10.0°. M_Z varies by less than 0.8° over temperature, and its total spread is 2.6°.

Figure 20 shows how the nulling resistance and $k = g_{m6}R_Z$ change with temperature for both designs.

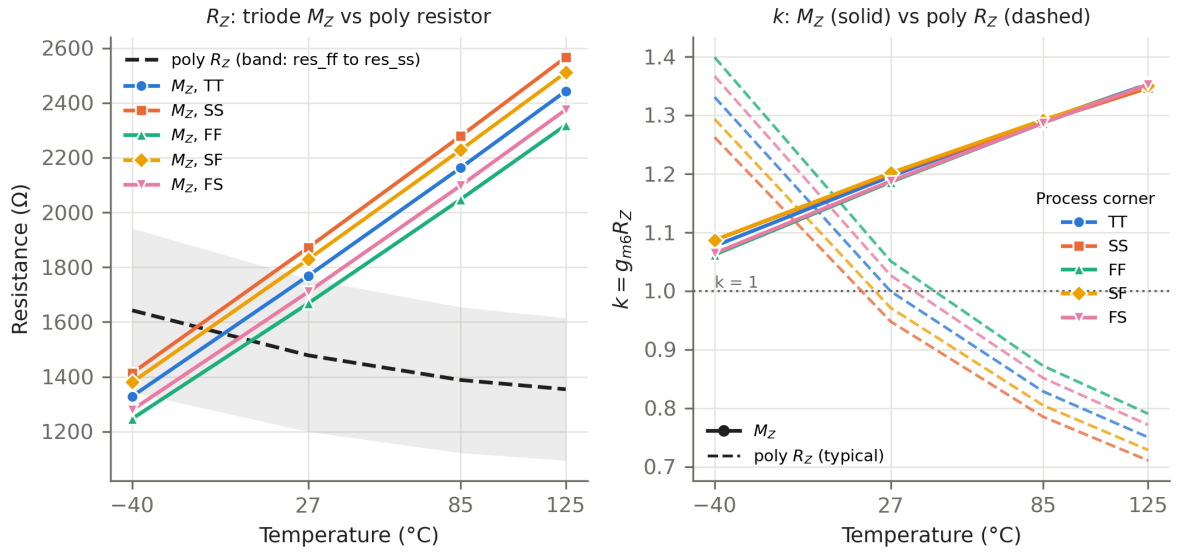


Figure 20: Left: on-resistance of the triode M_Z for the five process corners (solid lines) and the PDK poly resistor (dashed, band from `res_ff` to `res_ss`) versus temperature. Right: $k = g_{m6}R_Z$ for M_Z (solid lines) and the poly resistor (dashed). The dashed lines use the typical resistor; the `res_ss` and `res_ff` corners shift k by about ±20%, giving a full range of about 0.57 to 1.65. The on-resistance of M_Z rises with temperature as its mobility falls, while the poly resistance falls. As a result, k for M_Z is nearly the same at every process corner and rises from 1.077 at −40°C to 1.353 at 125°C , while k for the typical poly resistor falls from 1.26–1.40 to 0.71–0.79.

The two nulling elements respond to temperature in opposite directions (Fig. 20). The poly resistor has a negative temperature coefficient: at the typical corner R_Z falls from 1643 Ω at

−40 °C to 1354 Ω at 125 °C (Table 19). Since g_{m6} also falls with temperature, both terms of $k = g_{m6}R_Z$ decrease, and k drops from about 1.3 to 0.75 at TT. As k falls through 1, the zero moves from the left half-plane through infinity into the right half-plane, adding phase lag. With the nominal resistor, the phase margin therefore falls by 4.9° to 5.1° over temperature across the five process corners. The resistor corners change both the phase margin and its temperature drift: with `res_ss` (+20% sheet resistance) the drop is 5.3° to 5.6°, and with `res_ff` (−20%) it is 4.3° to 4.5°. Relative to the nominal resistor at the same process corner and temperature, the resistor corners shift the phase margin by about ±1.2° on average (±1.6° at most, at −40 °C). Together with the process corners, the phase margin spans 61.6° to 71.6° (10.0°), with the worst case at SF, 125 °C, `res_ff`.

The on-resistance of M_Z, $R_{MZ} = 1/[\mu_n C_{ox}(W/L)_Z V_{OV,Z}]$, depends on the same electron mobility as g_{m6}. As the mobility falls with temperature, R_{MZ} rises from 1329 Ω to 2443 Ω at TT while g_{m6} falls, so the two partly cancel. k varies by at most 2.3% across the five process corners at any temperature (worst case at −40 °C), and rises with temperature from 1.077 to 1.353 at TT, moving the zero further into the left half-plane. As a result, the phase margin of M_Z changes by less than 0.8° over temperature at any corner. Most of its total spread of 2.6° is between corners: SF is lowest and FS highest, because the PMOS input pair and the NMOS output stage move in opposite directions and M_Z tracks only the NMOS g_{m6}. This can be seen from Eq. (11), as phase margin depends on the ratio of $\frac{g_{m1}}{g_{m6}}$. Therefore, in the SF corner, $\frac{g_{m1}}{g_{m6}}$ will increase and lower the phase margin. In the FS corner, $\frac{g_{m1}}{g_{m6}}$ will decrease and increase the phase margin. M_Z also raises the worst-case gain margin from 20.3 dB to 29.7 dB.

Table 13 shows the phase margin by temperature, across all corners.

Table 13: Phase Margin by Temperature, Across All Corners (ngspice)

Temperature	Poly R_Z, Typical	Poly R_Z, `res_ff` to `res_ss`	M_Z
−40 °C	67.7° to 70.2°	66.1° to 71.6°	64.2° to 66.3°
27 °C	64.8° to 67.5°	63.6° to 68.7°	64.3° to 66.4°
85 °C	63.3° to 66.0°	62.2° to 67.0°	64.5° to 66.6°
125 °C	62.6° to 65.3°	61.6° to 66.3°	64.7° to 66.8°

Table 14 shows the summary for both designs.

Table 14: Summary of Corner Analysis for Both Designs (ngspice)

	Poly R_Z, `res_ff` to `res_ss`	M_Z
Phase Margin Spread Within a Corner	7.2° to 7.6°	0.3° to 0.7°
Phase Margin, Total Spread	10.0°	2.6°
Worst-Case Phase Margin	61.6°	64.2°
Gain Margin Range	20.3 dB to 39.5 dB	29.7 dB to 37.2 dB
Unity-Gain Frequency Range	28.4 MHz to 44.1 MHz	27.8 MHz to 43.0 MHz

Figure 21 shows the DC gain, unity-gain frequency, phase margin and gain margin of the M_Z design across all corners and temperatures.

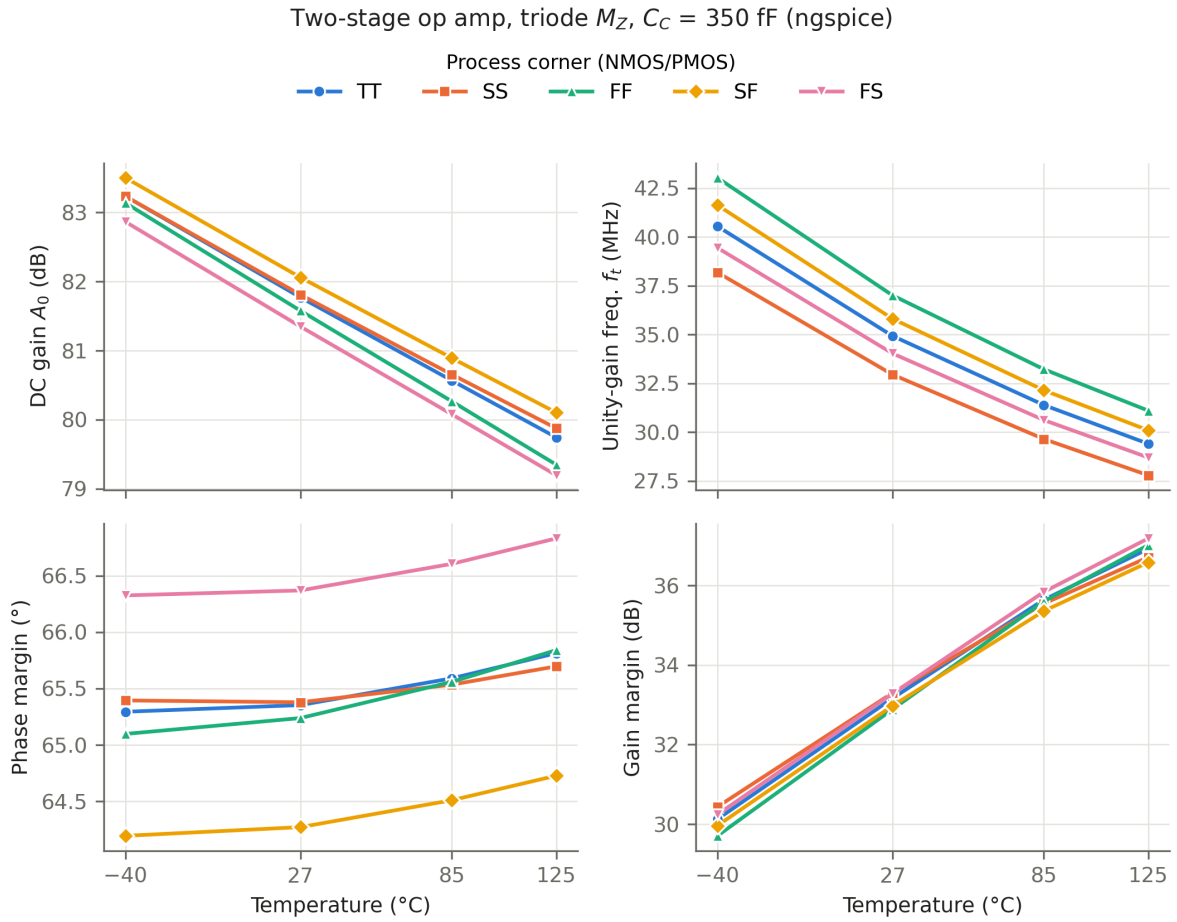


Figure 21: AC performance of the M_Z design across process corners and temperature (C_C = 350 fF, ngspice). $f_t \approx g_{m1}/(2\pi C_C)$ is set by the PMOS input pair, so the fast-PMOS corners (FF, SF) lie above the slow-PMOS corners (SS, FS). The phase margin is lowest at SF and highest at FS, where the PMOS input pair and the NMOS output stage move in opposite directions. Worst case: A_0 = 79.2 dB (FS, 125 °C), f_t = 27.8 MHz (SS, 125 °C), PM = 64.2° (SF, −40 °C) and GM = 29.7 dB (FF, −40 °C).

9.2 Monte Carlo

The corner analysis of Section 9.1 moves all NMOS devices together and all PMOS devices together. As a result, it does not show random mismatch between identical devices. Monte Carlo analysis is used to capture this effect. The analysis is run in ngspice-47 with the full PDK library (`sm141064.ngspice`), using the same inductor/capacitor bench as Section 6. The PDK provides two independent switches [3]:

- `sw_stat_global`: process variation shared by all devices of the same type (all NMOS move together, all PMOS move together), as in a corner analysis but drawn from a distribution.
- `sw_stat_mismatch`: independent variation of each device. The threshold-voltage mismatch follows Pelgrom's rule, $\sigma_{V_T} = 0.7071\, A_{VT}/\sqrt{L_{\text{eff}} W_{\text{eff}}}$, with A_{VT} = 7.148 mV µm (NMOS) and 6.66 mV µm (PMOS), $L_{\text{eff}} = L - 0.15\,\mu\text{m}$ and $W_{\text{eff}} = W + 0.1\,\mu\text{m}$.

Three Monte Carlo runs are performed, each with 500 trials at TT and 27 °C (further runs for the bias network are in Section 9.3). The netlist is given in Section B.4. The first two use the final design with the triode transistor M_Z: one with mismatch only, and one with global plus mismatch, which is the realistic case. The third run replaces M_Z with a resistor, for comparison

with the zero-cancelling resistor R_Z. An ideal resistor has a fixed value and would not vary in a Monte Carlo run, so the GF180MCU poly resistor from the PDK is used instead, which includes process variation. This run uses global plus mismatch variation. Table 15 summarizes all three runs, and Figs. 22 to 24 show the histograms.

Table 15: Monte Carlo Results, 500 Trials, TT, 27 °C (mean μ, standard deviation σ, and range)

	M_Z, Mismatch Only		M_Z, Global + Mismatch		PDK R_Z, Global + Mismatch	
	$\mu \pm \sigma$	Range	$\mu \pm \sigma$	Range	$\mu \pm \sigma$	Range
V_{OS} (mV)	0.05509 ± 8.96	-32.24 to 26.10	0.05591 ± 8.96	-32.44 to 25.54	0.329 ± 8.74	-27.70 to 23.44
DC Gain (dB)	81.77 ± 0.126	81.36 to 82.17	81.75 ± 0.460	80.37 to 83.18	81.76 ± 0.445	80.37 to 83.02
f_t (MHz)	34.93 ± 0.719	32.50 to 37.05	34.96 ± 0.893	32.59 to 38.51	35.65 ± 0.895	32.85 to 39.04
Phase Margin (°)	65.43 ± 0.683	63.59 to 68.08	65.72 ± 1.81	62.07 to 75.25	66.17 ± 0.577	64.59 to 67.80
$k = R_Z\, g_{m6}$	1.206 ± 0.0799	0.9888 to 1.521	1.250 ± 0.248	0.8272 to 2.768	0.9977 ± 0.0496	0.8442 to 1.176
R_Z (Ω)	1784 ± 119	1459 to 2226	1849 ± 366	1215 to 4048	1479 ± 71.0	1261 to 1736
Power (mW)	0.4270 ± 0.00734	0.4062 to 0.4547	0.4270 ± 0.00804	0.4082 to 0.4568	0.3996 ± 0.00666	0.3824 to 0.4216

Note: For M_Z, R_Z is the on-resistance $1/g_{ds}$ of the triode transistor. "PDK R_Z" replaces M_Z with the GF180MCU poly resistor (`ppolyf_u_1k`), sized $W = 2\,\mu$m, $L = 2.75\,\mu$m for a nominal $R_Z \approx 1.48\,$kΩ. Its sheet resistance, width and contact resistance vary with the PDK's global statistical parameters; no mismatch variation is applied to the resistor. All other devices are identical to the M_Z design. All three runs use the same random seed.

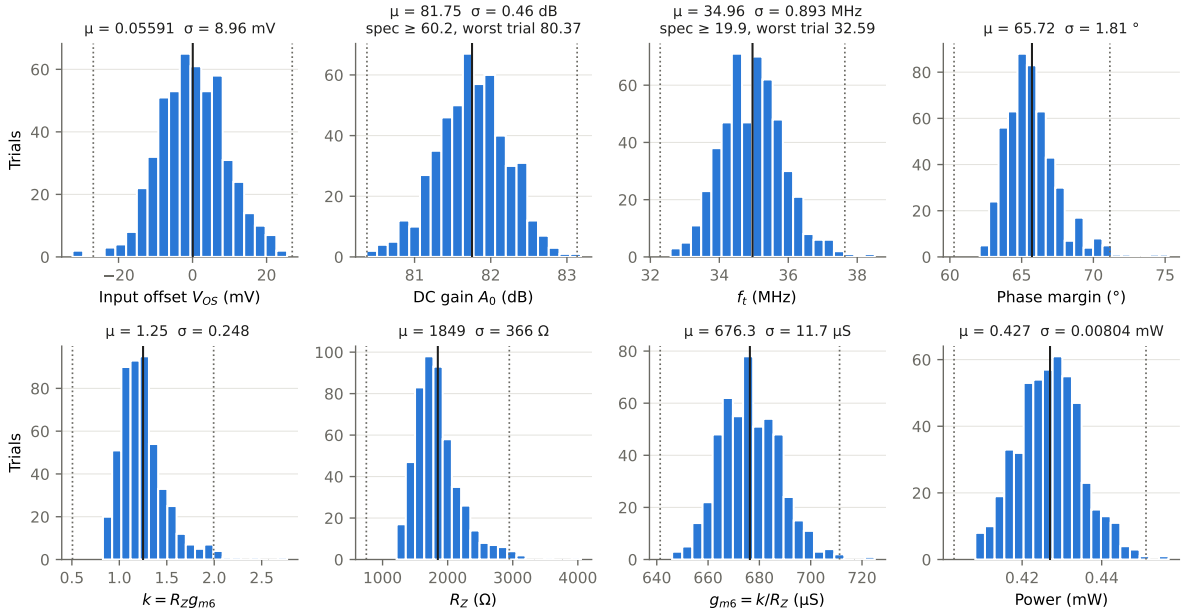


Figure 22: Monte Carlo histograms, triode M_Z, global plus mismatch variation, 500 trials, TT, 27 °C. The solid line marks the mean μ and the dotted lines $\mu \pm 3\sigma$.

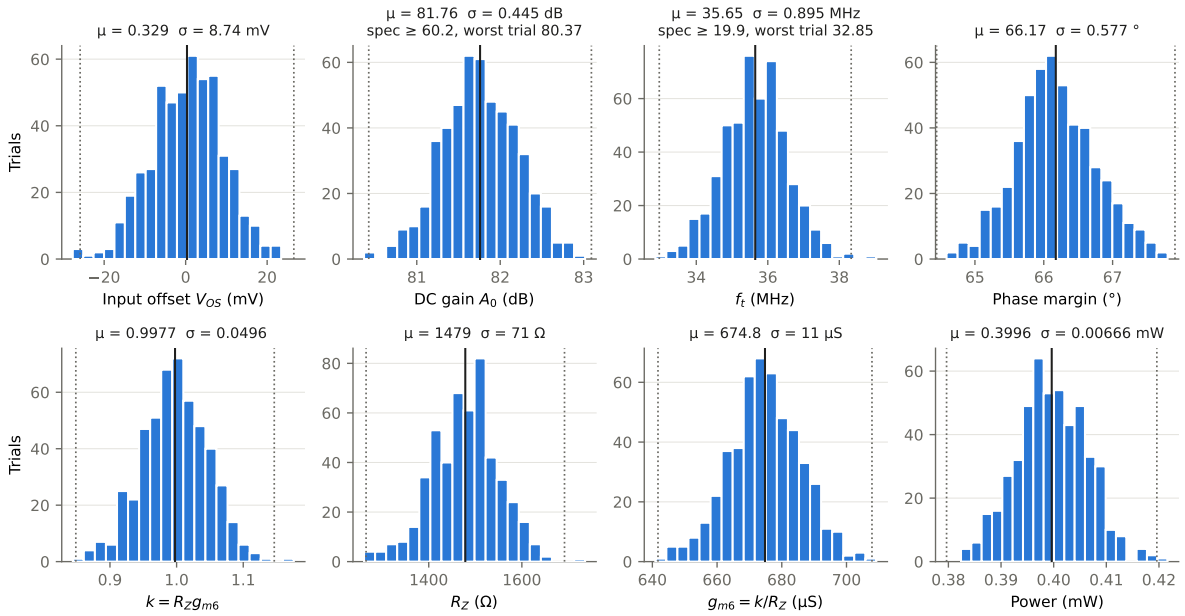


Figure 23: Monte Carlo histograms, PDK poly resistor R_Z, global plus mismatch variation, 500 trials, TT, 27 °C. The solid line marks the mean μ and the dotted lines $\mu \pm 3\sigma$.

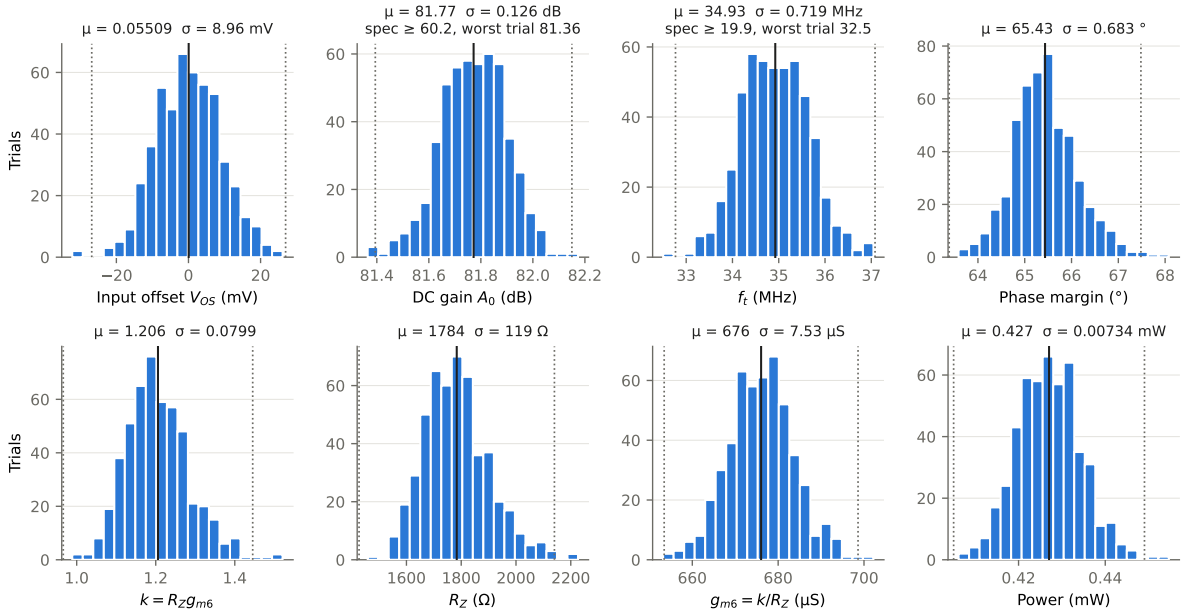


Figure 24: Monte Carlo histograms, triode M_Z, mismatch only, 500 trials, TT, 27 °C. The solid line marks the mean μ and the dotted lines $\mu \pm 3\sigma$.

Input Offset All three runs have a similar offset standard deviation of about 9 mV: 8.96 mV with mismatch only for M_Z, and 8.96 mV and 8.74 mV for M_Z and the PDK R_Z with global variation added. Replacing M_Z with a resistor does not change it, as neither carries DC current: they are in series with C_C, which is an open circuit at DC. The offset is therefore set by mismatch in the first stage in all three runs. In all three runs the mean offset is close to zero. This can be attributed to the condition from Eq. (1) to reduce systematic offset. Thus, the $\pm 3\sigma$ range for the global-plus-mismatch runs is about ± 26.88 mV for M_Z and ± 26.22 mV for the PDK R_Z.

Gain and Bandwidth All three runs have a mean DC gain of around 81.8 dB, matching closely with the 81.7 dB found in the AC analysis. For M_Z, global variation increases the standard deviation of the gain from 0.13 dB to 0.46 dB compared to mismatch only. With global plus mismatch variation, the lowest gain is 80.37 dB for both M_Z and the PDK R_Z. For M_Z, this equals its $\mu - 3\sigma$ value of 80.37 dB and the PDK R_Z is close to its value of 80.425 dB.

The unity-gain frequency has a similar spread in all three runs, with a standard deviation of 0.7 to 0.9 MHz, with M_Z and the PDK R_Z having a very similar standard deviation of 0.893 MHz and 0.895 MHz respectively, for global plus mismatch. The PDK R_Z gives a slightly higher mean (35.65 MHz against 34.96 MHz for M_Z). With global plus mismatch variation, the lowest unity-gain frequency is 32.59 MHz for M_Z, above its $\mu - 3\sigma$ value of 32.281 MHz, and 32.85 MHz for the PDK R_Z, just below its $\mu - 3\sigma$ value of 32.965 MHz.

Phase Margin With global plus mismatch variation, the mean phase margin is 65.72° for M_Z and 66.17° for the PDK R_Z, but the standard deviation is 1.81° for M_Z against 0.58° for the resistor. The phase margin depends on $k = R_Z g_{m6}$. The poly resistor has a small standard deviation of 71 Ω, therefore k stays between 0.84 and 1.18. The on-resistance of M_Z also has a higher standard deviation of 366 Ω. For M_Z, k varies from 0.83 to 2.77 and the phase margin follows it; the cause is examined in Section 9.3. The M_Z distribution is skewed towards higher phase margins (skewness +1.18), with a maximum of 75.25°. The lowest phase margin is 62.07° for M_Z and 64.59° for the PDK R_Z. From the corner analysis in Section 9.1, the triode implementation holds its phase margin better across temperature than the poly resistor. With M_Z, the phase margin changes by less than 0.8° from −40 to 125 °C across any corner, while with the poly resistor it changes by about 5°. Across all corners and temperatures, the phase margin spans 64.2° to 66.8° (a spread of 2.6°) with M_Z, compared with 61.6° to 71.6° (10°) for the poly resistor. g_{m6} and the poly resistor resistance fall as temperature increases, so k is expected to drift. The on-resistance of M_Z increases with temperature while g_{m6} decreases, so k stays above 1 and the zero remains in the left half-plane, which explains the smaller change in phase margin in Fig. 19. The Monte Carlo results show the opposite trade-off at a fixed temperature. The poly resistor gives a smaller phase-margin spread from chip to chip (0.58° against 1.81°).

9.3 Spread of k in the M_Z Design

The Monte Carlo results show a much larger spread in k for the M_Z design ($\sigma_k = 0.248$) than for the poly resistor ($\sigma_k = 0.0496$). Since g_{m6} varies by less than 2% in both designs, the difference comes from the resistor itself.

The poly resistor depends mainly on its sheet resistance, which varies by about 5%. M_Z, by contrast, is a transistor in triode with $R_{M_Z} = 1/[\mu_n C_{ox}(W/L)_Z V_{\mathrm{OVZ}}]$, and it operates at a small overdrive ($V_{\mathrm{OVZ}} \approx 0.2\,\mathrm{V}$). Since $R_{M_Z} \propto 1/V_{\mathrm{OVZ}}$, a small change in overdrive gives $\Delta R_{M_Z}/R_{M_Z} \approx -\Delta V_{\mathrm{OVZ}}/V_{\mathrm{OVZ}}$, so with $V_{\mathrm{OVZ}} \approx 0.2\,\mathrm{V}$ each millivolt of error changes R_{M_Z} by about $1/200 = 0.5\%$.

The bias network (M_P, M_A, M_B) is designed to cancel this: M_A copies V_{GS6} and M_B copies $V_{GS,Z}$, so threshold shifts common to all devices cancel and V_{OVZ} stays fixed. The cancellation fails because the bias devices are narrow (1.344 and 5.48 µm), while M_6, M_Z and M_8 are wide ($\geq 10\,\mu\mathrm{m}$). The GF180MCU model assigns different global threshold variation to these two width ranges, so a process shift moves the bias devices more than the devices they are meant to track. The narrow devices also have more random mismatch. Together these leave tens of millivolts of error on V_{OVZ}, which becomes a $\sim 20\%$ spread in R_{M_Z}.

The GF180MCU model divides each device type into width and length bins, and assigns each bin its own global threshold variation. For $L = 0.56\,\mu\mathrm{m}$, the relevant bins are listed in Table 16. M_6, M_Z, M_7 and M_8 lie in the wide bin, while the bias devices M_A, M_B and M_P lie in the narrow bin, which varies more.

Table 16: Global threshold variation (1σ) of the GF180MCU model bins used in the two-stage op amp ($0.5 \le L < 1.2\,\mu$m).

Bin	Devices	NMOS	PMOS
$1.2 \le W < 10\,\mu$m	M_A, M_B ($1.344\,\mu$m), M_P ($5.48\,\mu$m)	25 mV	30 mV
$W \ge 10\,\mu$m	M_6, M_Z, M_7, M_8	15 mV	18 mV

A global shift therefore moves the bias devices differently from the devices they are meant to track, so the shift no longer cancels in the bias stack.

To confirm global variation is reduced, the bias devices, M_A, M_B and M_P, were scaled up by $8\times$ (to 10.75, 10.75 and $43.84\,\mu$m), which places them in the wide bin with M_6, M_Z and M_8. Monte Carlo runs were performed for the scaled circuit with 500 trials. Table 17 summarizes the Monte Carlo runs for the original design with the triode resistor, the poly R_Z, and the scaled triode resistor design ($8\times$).

Table 17: Monte Carlo spread of k and phase margin for the two-stage op amp (500 trials each, TT, 27 °C, same seed). The $8\times$ design scales M_A, M_B and M_P by $8\times$ so that all tracking devices share the same model width bin.

Design	Variation	k ($\mu \pm \sigma$)	k range	$\sigma(R)/\mu(R)$	PM ($\mu \pm \sigma$)	PM range
M_Z, original	Global + mismatch	1.250 ± 0.248	0.83–2.77	19.8%	$65.72 \pm 1.81°$	62.1–75.2°
	Global only	1.241 ± 0.238	0.81–3.09	19.2%	$65.65 \pm 1.68°$	62.3–77.5°
	Mismatch only	1.206 ± 0.080	0.99–1.52	6.7%	$65.43 \pm 0.68°$	63.6–68.1°
M_Z, $8\times$ bias	Global + mismatch	1.265 ± 0.043	1.14–1.42	3.7%	$65.70 \pm 0.57°$	64.1–67.6°
	Global only	1.262 ± 0.006	1.24–1.29	1.5%	$65.67 \pm 0.22°$	65.1–66.4°
	Mismatch only	1.265 ± 0.042	1.14–1.42	3.3%	$65.70 \pm 0.52°$	64.3–67.6°
Poly R_Z	Global + mismatch	0.998 ± 0.050	0.84–1.18	4.8%	$66.17 \pm 0.58°$	64.6–67.8°

In the global plus mismatch simulations for the triode implementation, the spread in k fell from $\sigma_k \approx 0.248$ to 0.043, and σ_{PM} from 1.81° to 0.57° (Fig. 25), matching the poly-resistor design closely, at the cost of about $187\,\mu$W of extra bias power.

Scaling also reduces the random mismatch of M_A and M_B, so the global-only runs separate the two effects. With mismatch disabled, the spread in k fell from $\sigma_k \approx 0.238$ to 0.006; therefore, the improvement comes from global tracking rather than reduced mismatch. With mismatch only, σ_k fell from 0.080 to 0.042. In the $8\times$ design, almost all of the remaining spread in k and phase margin comes from mismatch ($\sqrt{0.006^2 + 0.042^2} \approx 0.043$), whereas in the original design it comes from global variation.

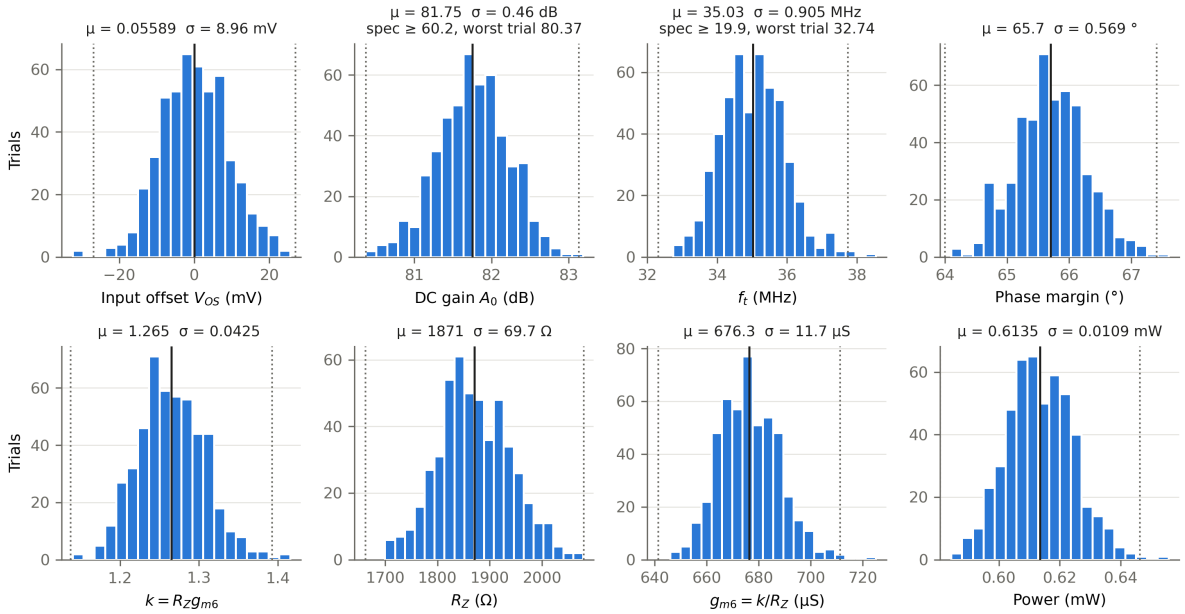


Figure 25: Monte Carlo distributions of the two-stage op amp with M_Z and its bias devices M_A, M_B and M_P scaled by 8× (500 trials, global plus mismatch, TT, 27 °C, same seed as the original run). Solid lines mark the mean and dotted lines $\mu \pm 3\sigma$. R_Z here is $R_{M_Z} = 1/g_{ds,Z}$, and $g_{m6} = k/R_{M_Z}$ is computed per trial. Compared with the original bias, the spread in R_{M_Z} falls from 19.8% to 3.7% and in k from $\sigma = 0.248$ to 0.043, and the phase margin narrows to $65.70 \pm 0.57°$.

10 Conclusion

A two-stage CMOS op amp was designed in the GF180MCU process using a V_{OV}-based sizing method. Without compensation, the phase margin is $-8.2°$. A compensation capacitor of 233.8 fF was designed for the 60° target and gives 56.9° in simulation. However, the step response showed that the output stage then limits the rising slew rate. Requiring the first stage to limit slewing sets $C_{\mathrm{C}} \geq 330$ fF, and 350 fF was used for margin.

The right-half-plane zero is removed with a series resistance of $1/g_{\mathrm{m6}} = 1479\,\Omega$, implemented either as a resistor or as a triode NMOS M_{Z} sized for $k = R_{\mathrm{Z}} g_{\mathrm{m6}} = 1.2$. At TT and 27 °C, the M_{Z} design achieves $f_{\mathrm{t}} = 34.9$ MHz, a phase margin of 64.9°, a gain margin of 32.4 dB and a CMRR of 83.1 dB, for 0.427 mW. The resistor design reaches 65.9° and 27.0 dB at 0.400 mW; the extra 27 μW is the gate-bias branch of M_{Z}.

Across corners and temperature, the phase margin with M_{Z} stays between 64.2° and 66.8° and changes by less than 0.8° within any corner. The phase margin for the poly resistor stays between 61.6° and 71.6° and has a spread of 7.2° to 7.6° within each corner.

In the Monte Carlo simulations, the poly resistor gives the tighter chip-to-chip phase margin spread ($\sigma = 0.58°$ against 1.81°). The input offset has a standard deviation of 8.96 mV with M_{Z} and a standard deviation of 8.74 mV for the poly resistor, and is set by the first stage in both designs.

For M_Z design, the standard deviation of k and R_{MZ} can be reduced by making the bias transistors of a similar geometry as M_6, M_Z, and M_8. This is done by increasing the width of the bias transistors to be in the same model bin as M_6, M_Z, and M_8. Scaling M_A, M_B and M_P by 8× reduces σ_k from 0.248 to 0.043 and the phase-margin spread to 0.57°, matching the poly resistor, at the cost of 187 μW. A global-only run shows that without mismatch, σ_k falls from 0.238 to 0.006, confirming that the improvement comes from global tracking.

References

[1] A. S. Sedra and K. C. Smith, *Microelectronic Circuits*, 7th ed. Oxford University Press, 2015.

[2] B. Razavi, *Design of Analog CMOS Integrated Circuits*, 2nd ed. McGraw-Hill Education, 2017.

[3] GlobalFoundries, "GF180MCU open-source PDK," model release 2024-01-01.

A Full Corner Data

R_Z is the on-resistance $1/g_{ds}$ of M_Z or the value of the poly resistor, and $k = g_{m6}R_Z$.

Table 18: Corner Data, Triode M_Z

Corner	T (°C)	R_Z (Ω)	k	A_0 (dB)	f_t (MHz)	PM (°)	f_{180} (MHz)	GM (dB)
TT	−40	1329	1.077	83.2	40.5	65.3	373	30.1
	27	1769	1.196	81.8	34.9	65.4	389	33.2
	85	2163	1.291	80.6	31.4	65.6	418	35.6
	125	2443	1.353	79.7	29.4	65.8	436	36.9
SS	−40	1414	1.086	83.2	38.2	65.4	359	30.4
	27	1872	1.200	81.8	33.0	65.4	371	33.3
	85	2279	1.289	80.7	29.6	65.5	392	35.5
	125	2566	1.346	79.9	27.8	65.7	405	36.7
FF	−40	1247	1.062	83.1	43.0	65.1	385	29.7
	27	1668	1.186	81.6	37.0	65.2	404	32.9
	85	2048	1.286	80.3	33.2	65.6	440	35.6
	125	2319	1.353	79.3	31.1	65.8	464	37.0
SF	−40	1380	1.087	83.5	41.6	64.2	372	30.0
	27	1829	1.201	82.1	35.8	64.3	387	33.0
	85	2228	1.292	80.9	32.1	64.5	413	35.4
	125	2511	1.350	80.1	30.1	64.7	428	36.6
FS	−40	1279	1.064	82.9	39.4	66.3	373	30.2
	27	1710	1.187	81.3	34.0	66.4	389	33.3
	85	2098	1.287	80.1	30.6	66.6	420	35.8
	125	2375	1.352	79.2	28.7	66.8	440	37.2

Table 19: Corner Data, PDK Poly R_Z, Typical

Corner	T (°C)	R_Z (Ω)	k	A_0 (dB)	f_t (MHz)	PM (°)	f_{180} (MHz)	GM (dB)
TT	−40	1643	1.331	83.2	41.5	69.0	600	36.0
	27	1478	0.999	81.8	35.7	66.2	278	27.1
	85	1388	0.828	80.6	32.0	64.7	204	24.0
	125	1354	0.750	79.7	30.0	64.0	177	22.8
SS	−40	1643	1.262	83.2	39.0	68.5	489	34.1
	27	1478	0.948	81.8	33.6	65.9	246	26.2
	85	1388	0.785	80.7	30.3	64.4	185	23.3
	125	1354	0.711	79.9	28.4	63.6	161	22.2
FF	−40	1643	1.399	83.1	44.0	69.3	730	37.6
	27	1478	1.051	81.6	37.8	66.5	315	28.3
	85	1388	0.872	80.3	33.9	65.0	224	24.6
	125	1354	0.790	79.3	31.7	64.3	193	23.3
SF	−40	1643	1.293	83.5	42.6	67.7	547	34.7
	27	1478	0.971	82.1	36.6	64.8	261	26.1
	85	1388	0.805	80.9	32.9	63.3	194	23.1
	125	1354	0.729	80.1	30.8	62.6	169	22.0
FS	−40	1643	1.366	82.9	40.3	70.2	651	37.1
	27	1478	1.026	81.3	34.7	67.5	294	28.2
	85	1388	0.851	80.1	31.2	66.0	213	24.8
	125	1354	0.771	79.2	29.2	65.3	184	23.5

Table 20: Corner Data, PDK Poly R_Z, `res_ss` (+20% Sheet Resistance)

Corner	T (°C)	R_Z (Ω)	k	A_0 (dB)	f_t (MHz)	PM (°)	f_{180} (MHz)	GM (dB)
TT	−40	1942	1.573	83.2	41.5	70.5	852	39.2
	27	1757	1.188	81.8	35.7	67.4	376	31.9
	85	1654	0.987	80.6	32.0	65.7	242	26.8
	125	1614	0.894	79.7	30.0	64.9	202	25.0
SS	−40	1942	1.492	83.2	39.1	69.9	734	38.7
	27	1757	1.126	81.8	33.7	67.0	317	30.2
	85	1654	0.936	80.7	30.3	65.4	215	25.8
	125	1614	0.847	79.9	28.4	64.5	182	24.2
FF	−40	1942	1.653	83.1	44.1	70.9	964	39.3
	27	1757	1.249	81.6	37.8	67.8	454	33.7
	85	1654	1.039	80.3	33.9	66.1	275	27.9
	125	1614	0.942	79.3	31.7	65.3	225	25.8
SF	−40	1942	1.529	83.5	42.7	69.2	809	38.7
	27	1757	1.154	82.1	36.7	66.1	346	30.5
	85	1654	0.959	80.9	32.9	64.4	228	25.8
	125	1614	0.868	80.1	30.8	63.6	192	24.1
FS	−40	1942	1.615	82.9	40.3	71.6	888	39.5
	27	1757	1.220	81.3	34.7	68.7	408	33.2
	85	1654	1.015	80.1	31.2	67.0	257	27.8
	125	1614	0.919	79.2	29.2	66.3	213	25.9

Table 21: Corner Data, PDK Poly R_Z, `res_ff` (-20% Sheet Resistance)

Corner	T (°C)	R_Z (Ω)	k	A_0 (dB)	f_t (MHz)	PM (°)	f_{180} (MHz)	GM (dB)
TT	−40	1343	1.088	83.2	41.4	67.4	375	29.3
	27	1200	0.811	81.8	35.7	65.0	226	23.8
	85	1122	0.670	80.6	32.0	63.7	178	21.8
	125	1095	0.606	79.7	30.0	63.0	159	21.0
SS	−40	1343	1.032	83.2	39.0	67.1	327	28.1
	27	1200	0.769	81.8	33.6	64.7	206	23.3
	85	1122	0.635	80.7	30.3	63.4	164	21.4
	125	1095	0.574	79.9	28.4	62.7	147	20.6
FF	−40	1343	1.144	83.1	44.0	67.7	436	30.7
	27	1200	0.853	81.6	37.8	65.2	249	24.4
	85	1122	0.705	80.3	33.9	63.9	194	22.2
	125	1095	0.639	79.3	31.7	63.3	171	21.3
SF	−40	1343	1.058	83.5	42.6	66.1	350	28.2
	27	1200	0.788	82.1	36.6	63.6	216	23.0
	85	1122	0.651	80.9	32.9	62.2	171	21.1
	125	1095	0.589	80.1	30.8	61.6	153	20.3
FS	−40	1343	1.118	82.9	40.2	68.7	401	30.5
	27	1200	0.833	81.3	34.7	66.3	237	24.6
	85	1122	0.688	80.1	31.2	65.0	186	22.5
	125	1095	0.623	79.2	29.2	64.4	165	21.7

B SPICE Netlists

B.1 Two-Stage Op-Amp and AC Test Bench

The netlist below was simulated in LTspice with the GF180MCU models at TT and 27 °C.

```
* Two-stage Miller op-amp (M_Z nulling), GF180MCU 3.3 V, AC test bench (TT, 27 C)
* LFB/CFB close the loop at DC (Vout = 1.65 V) and open it at AC,
* so V(vo)/V(vs) = -A.

* Supplies
V1 VSS 0 0
V2 VDD 0 3.3

* op-amp
M5 N003 N001 VDD VDD pmos_3p3 l={L5} w={W5} ad={W5*E} as={W5*E} pd={2*W5+2*E} ps={2*W5+2*E} m=1
M1 N007 v- N003 VDD pmos_3p3 l={L1} w={W1} ad={W1*E} as={W1*E} pd={2*W1+2*E} ps={2*W1+2*E} m=1
M2 N005 v+ N003 VDD pmos_3p3 l={L2} w={W2} ad={W2*E} as={W2*E} pd={2*W2+2*E} ps={2*W2+2*E} m=1
M3 N007 N007 VSS VSS nmos_3p3 l={L3} w={W3} ad={W3*E} as={W3*E} pd={2*W3+2*E} ps={2*W3+2*E} m=1
M4 N005 N007 VSS VSS nmos_3p3 l={L4} w={W4} ad={W4*E} as={W4*E} pd={2*W4+2*E} ps={2*W4+2*E} m=1
M7 Vo N001 VDD VDD pmos_3p3 l={L7} w={W7} ad={W7*E} as={W7*E} pd={2*W7+2*E} ps={2*W7+2*E} m=1
M6 Vo N005 VSS VSS nmos_3p3 l={L6} w={W6} ad={W6*E} as={W6*E} pd={2*W6+2*E} ps={2*W6+2*E} m=1
CL Vo VSS 1pF

* Miller compensation with triode nulling device M_Z
CC Vo N006 350f
MZ N005 N002 N006 VSS nmos_3p3 l={LZ} w={WZ} ad={WZ*E} as={WZ*E} pd={2*WZ+2*E} ps={2*WZ+2*E} m=1

* Bias circuit
IREF N001 VSS 20e-6
M8 N001 N001 VDD VDD pmos_3p3 l={L8} w={W8} ad={W8*E} as={W8*E} pd={2*W8+2*E} ps={2*W8+2*E} m=1
MP N002 N001 VDD VDD pmos_3p3 l={LP} w={WP} ad={WP*E} as={WP*E} pd={2*WP+2*E} ps={2*WP+2*E} m=1
MB N002 N002 N004 VSS nmos_3p3 l={LB} w={WB} ad={WB*E} as={WB*E} pd={2*WB+2*E} ps={2*WB+2*E} m=1
MA N004 N004 VSS VSS nmos_3p3 l={LA} w={WA} ad={WA*E} as={WA*E} pd={2*WA+2*E} ps={2*WA+2*E} m=1

* Test bench
vs N008 0 0 AC 1
CFB v- N008 1e6
LFB Vo v- 1e6
V3 v+ 0 1.65
```

```
* Models and sizes
.inc ./gf180_spice_models_20240101/cmos_tt.txt
.param E=0.44e-6
.param W1=6.85e-6 L1=0.56e-6
.param W2=6.85e-6 L2=0.56e-6
.param W3=1.68e-6 L3=0.56e-6
.param W4=1.68e-6 L4=0.56e-6
.param W5=13.7e-6 L5=0.56e-6
.param W6=13.44e-6 L6=0.56e-6
.param W7=54.8e-6 L7=0.56e-6
.param W8=13.7e-6 L8=0.56e-6
.param WP=5.48e-6 LP=0.56e-6
.param WA=1.344e-6 LA=0.56e-6
.param WB=1.344e-6 LB=0.56e-6
.param WZ=10.7e-6 LZ=0.56e-6

* Analysis and measurements
.ac dec 20 100 10G
.options meascplxfmt=cartesian
.meas AC a0 MAX mag(V(vo))
.meas AC a0db PARAM 20*log10(a0)
.meas AC f3db WHEN mag(V(vo))=a0/sqrt(2) FALL=1
.meas AC ft WHEN mag(V(vo))=1
.meas AC ph FIND ph(-V(vo)) WHEN mag(V(vo))=1
.meas AC pm PARAM 180+ph
.meas AC f180 WHEN Im(V(vo))=0 FALL=1
.meas AC g180 FIND mag(V(vo)) WHEN Im(V(vo))=0 FALL=1
.meas AC gmdb PARAM -20*log10(g180)
.end
```

B.2 Transient Test Bench

The slew rate, rise and fall times and overshoot are simulated in LTspice with the op amp connected as a unity-gain buffer (the output drives the gate of M_1) and driven through 1 kΩ by a 1 V input step. The listing below is the M_Z design.

```
* Two-stage Miller op-amp (M_Z nulling), unity-gain step response (TT, 27 C)
* Output node v- drives the gate of M1 (unity-gain buffer)

* Supplies
V1 VSS 0 0
V2 VDD 0 3.3

* op-amp
M5 N004 N002 VDD VDD pmos_3p3 l={L5} w={W5} ad={W5*E} as={W5*E} pd={2*W5+2*E} ps={2*W5+2*E} m=1
M1 N008 v- N004 VDD pmos_3p3 l={L1} w={W1} ad={W1*E} as={W1*E} pd={2*W1+2*E} ps={2*W1+2*E} m=1
M2 N006 v+ N004 VDD pmos_3p3 l={L2} w={W2} ad={W2*E} as={W2*E} pd={2*W2+2*E} ps={2*W2+2*E} m=1
M3 N008 N008 VSS VSS nmos_3p3 l={L3} w={W3} ad={W3*E} as={W3*E} pd={2*W3+2*E} ps={2*W3+2*E} m=1
M4 N006 N008 VSS VSS nmos_3p3 l={L4} w={W4} ad={W4*E} as={W4*E} pd={2*W4+2*E} ps={2*W4+2*E} m=1
M7 v- N002 VDD VDD pmos_3p3 l={L7} w={W7} ad={W7*E} as={W7*E} pd={2*W7+2*E} ps={2*W7+2*E} m=1
M6 v- N006 VSS VSS nmos_3p3 l={L6} w={W6} ad={W6*E} as={W6*E} pd={2*W6+2*E} ps={2*W6+2*E} m=1
CL v- VSS 1pF

* Miller compensation with triode nulling device M_Z
CC v- N007 350fF
MZ1 N006 N003 N007 VSS nmos_3p3 l={LZ} w={WZ} ad={WZ*E} as={WZ*E} pd={2*WZ+2*E} ps={2*WZ+2*E} m=1

* Bias circuit
IREF1 N002 VSS 20e-6
M8 N002 N002 VDD VDD pmos_3p3 l={L8} w={W8} ad={W8*E} as={W8*E} pd={2*W8+2*E} ps={2*W8+2*E} m=1
MP N003 N002 VDD VDD pmos_3p3 l={L10} w={W10} ad={W10*E} as={W10*E} pd={2*W10+2*E} ps={2*W10+2*E} m=1
MB N003 N003 N005 VSS nmos_3p3 l={L11} w={W11} ad={W11*E} as={W11*E} pd={2*W11+2*E} ps={2*W11+2*E} m=1
MA N005 N005 VSS VSS nmos_3p3 l={L12} w={W12} ad={W12*E} as={W12*E} pd={2*W12+2*E} ps={2*W12+2*E} m=1

* Test bench: 1 V step, 0.8 V to 1.8 V
V4 N001 0 PULSE(0.8 1.8 0.1u 100p 100p 200n)
R1 v+ N001 1000

* Models and sizes
.inc ./gf180_spice_models_20240101/cmos_tt.txt
.param E=0.44e-6
```

```
.param W1=6.85e-6 L1=0.56e-6
.param W2=6.85e-6 L2=0.56e-6
.param W3=1.68e-6 L3=0.56e-6
.param W4=1.68e-6 L4=0.56e-6
.param W5=13.7e-6 L5=0.56e-6
.param W6=13.44e-6 L6=0.56e-6
.param W7=54.8e-6 L7=0.56e-6
.param W8=13.7e-6 L8=0.56e-6
.param W10=5.48e-6 L10=0.56e-6
.param W11=1.344e-6 L11=0.56e-6
.param W12=1.344e-6 L12=0.56e-6
.param WZ=10.7e-6 LZ=0.56e-6

* Analysis and measurements
.tran 0 600n 0 50p
* overshoot / undershoot
.meas tran vlo AVG V(v-) FROM 80n TO 100n
.meas tran vhi AVG V(v-) FROM 280n TO 300n
.meas tran vpk MAX V(v-) FROM 100n TO 300n
.meas tran vmin MIN V(v-) FROM 300n TO 500n
.meas tran os_mV PARAM (vpk-vhi)*1e3
.meas tran os_pct PARAM 100*(vpk-vhi)/(vhi-vlo)
.meas tran us_mV PARAM (vlo-vmin)*1e3
.meas tran us_pct PARAM 100*(vlo-vmin)/(vhi-vlo)
* 10-90% edges and slew rate (V/us)
.meas tran trise TRIG V(v-)=0.9 RISE=1 TARG V(v-)=1.7 RISE=1
.meas tran tfall TRIG V(v-)=1.7 FALL=1 TARG V(v-)=0.9 FALL=1
.meas tran sr_rise PARAM 0.8e-6/trise
.meas tran sr_fall PARAM 0.8e-6/tfall
* M6 gate voltage, M6 current and M_Z gate voltage during slewing
.meas tran vg6min MIN V(n006) FROM 100n TO 200n
.meas tran vg6max MAX V(n006) FROM 300n TO 400n
.meas tran idmin MIN Id(M6) FROM 100n TO 200n
.meas tran idmax MAX Id(M6) FROM 300n TO 400n
.meas tran vgzmax MAX V(n003) FROM 100n TO 500n
.meas tran vgzmin MIN V(n003) FROM 100n TO 500n
.end
```

The step-response figures in Section 7 use the ideal-resistor design, which replaces M_Z and its bias branch (MP, MB, MA) with an ideal resistor in series with C_C. Node `N006` (drains of M2 and M4, gate of M6) is named `VG6`. The first run uses $C_C = 233.8\,\text{fF}$ and the second $C_C = 350\,\text{fF}$. The lines that differ are:

```
* Miller compensation with ideal nulling resistor R_z = 1/gm6
Rz v- nz 1479
CC nz VG6 350f ; 233.8f for the first run
* N006 renamed VG6
* Delete: MZ1, MP, MB, MA, and the vgzmax / vgzmin measures
M2 VG6 v+ N004 VDD pmos_3p3 l={L2} w={W2} ad={W2*E} as={W2*E} pd={2*W2+2*E} ps={2*W2+2*E} m=1
M4 VG6 N008 VSS VSS nmos_3p3 l={L4} w={W4} ad={W4*E} as={W4*E} pd={2*W4+2*E} ps={2*W4+2*E} m=1
M6 v- VG6 VSS VSS nmos_3p3 l={L6} w={W6} ad={W6*E} as={W6*E} pd={2*W6+2*E} ps={2*W6+2*E} m=1
.meas tran vg6min MIN V(vg6) FROM 100n TO 200n
.meas tran vg6max MAX V(vg6) FROM 300n TO 400n
```

B.3 Corner Set-Up (ngspice)

The corner and temperature runs use ngspice-47 with the GF180MCU library. Each run selects one process corner with the `.lib` line (`typical`, `ff`, `ss`, `sf` or `fs`) and loops over the four temperatures in `.control`. The label in `set corner` only names the corner in the results file. The listing below is the M_Z design.

```
* Two-stage op amp, triode M_Z: process corners x temperature (ngspice)
* ---- CORNER: set the .lib section here (typical, ff, ss, sf, fs) and the label in .control ----
.lib sm141064.ngspice typical
.param sw_stat_global=0 sw_stat_mismatch=0
.param mc_skew=3 res_mc_skew=3 cap_mc_skew=3 fnoicor=0
.param E=0.44u
.param L0=0.56u
.param W1=6.85u W3=1.68u W5=13.7u W6=13.44u W7=54.8u W8=13.7u
```

```
.param WP=5.48u WA=1.344u WB=1.344u WZ=10.7u
.param CCV=350f

V1 VSS 0 0
V2 VDD 0 3.3

* ---- bias network ----
IREF N001 VSS 20u
XM8 N001 N001 VDD VDD pmos_3p3 w={W8} l={L0} ad={W8*E} as={W8*E} pd={2*W8+2*E} ps={2*W8+2*E}
XMP N002 N001 VDD VDD pmos_3p3 w={WP} l={L0} ad={WP*E} as={WP*E} pd={2*WP+2*E} ps={2*WP+2*E}
XMB N002 N002 N004 VSS nmos_3p3 w={WB} l={L0} ad={WB*E} as={WB*E} pd={2*WB+2*E} ps={2*WB+2*E}
XMA N004 N004 VSS VSS nmos_3p3 w={WA} l={L0} ad={WA*E} as={WA*E} pd={2*WA+2*E} ps={2*WA+2*E}

* ---- core (M1 = inverting, M2 = non-inverting input) ----
XM5 N003 N001 VDD VDD pmos_3p3 w={W5} l={L0} ad={W5*E} as={W5*E} pd={2*W5+2*E} ps={2*W5+2*E}
XM1 N007 vm N003 VDD pmos_3p3 w={W1} l={L0} ad={W1*E} as={W1*E} pd={2*W1+2*E} ps={2*W1+2*E}
XM2 N005 vp N003 VDD pmos_3p3 w={W1} l={L0} ad={W1*E} as={W1*E} pd={2*W1+2*E} ps={2*W1+2*E}
XM3 N007 N007 VSS VSS nmos_3p3 w={W3} l={L0} ad={W3*E} as={W3*E} pd={2*W3+2*E} ps={2*W3+2*E}
XM4 N005 N007 VSS VSS nmos_3p3 w={W3} l={L0} ad={W3*E} as={W3*E} pd={2*W3+2*E} ps={2*W3+2*E}
XM7 Vo N001 VDD VDD pmos_3p3 w={W7} l={L0} ad={W7*E} as={W7*E} pd={2*W7+2*E} ps={2*W7+2*E}
XM6 Vo N005 VSS VSS nmos_3p3 w={W6} l={L0} ad={W6*E} as={W6*E} pd={2*W6+2*E} ps={2*W6+2*E}
* Miller compensation with triode nulling device M_Z
CC Vo N006 {CCV}
XMZ N005 N002 N006 VSS nmos_3p3 w={WZ} l={L0} ad={WZ*E} as={WZ*E} pd={2*WZ+2*E} ps={2*WZ+2*E}
CL Vo VSS 1p

* ---- bench: DC feedback through L, AC drive into v- through C ----
Vcm vp 0 dc 1.65
LFB Vo vm 1e6
CFB vm vs 1e6
Vs vs 0 dc 0 ac 1

.control
set units = degrees
* ---- CORNER label for the results file (match the .lib line above) ----
set corner = TT
foreach tc -40 27 85 125
  reset
  option temp = $tc
  save all @m.xm6.m0[gm] @m.xmz.m0[gds]
  op
  * k = gm6 / gds(MZ) = gm6 * R_MZ
  let k = @m.xm6.m0[gm] / @m.xmz.m0[gds]
  let rmz = 1/@m.xmz.m0[gds]
  set s_rmz = $&rmz
  set s_k = $&k
  ac dec 100 100 10g
  meas ac a0 max vdb(vo)
  meas ac ft when vdb(vo)=0
  let avec = -1*v(vo)
  let pha = cph(avec)
  meas ac ph find pha when vdb(vo)=0
  meas ac f180 when pha=-180
  meas ac g180 find vdb(vo) when pha=-180
  let pm = 180 + ph
  let gm = -g180
  echo "$corner $tc mz $s_rmz $s_k $&a0 $&ft $&pm $&f180 $&gm" >> ts_corners_results.txt
  destroy all
end
quit
.endc
.end
```

The poly-resistor run replaces M_Z and its bias branch (MP, MB, MA) with the `ppolyf_u_1k` resistor ($2 \times 2.75\,\mu$m), written out from the PDK's own equations as in the Monte Carlo set-up (Section B.4), and adds a loop over the three resistor corners. The parameter `rz_rc` sets the sheet resistance: 0 for typical, +0.2 for `res_ss` and −0.2 for `res_ff`, the same values as the PDK's resistor corner sections. The lines that differ are:

```
* Two-stage op amp, PDK poly R_Z: process corners x temperature x resistor corner (ngspice)

* ---- R_Z = GF180 ppolyf_u_1k, W = 2 um, L = 2.75 um, from the PDK's own equations ----
```

```
.param rz_T=27 rz_rc=0
.param rz_dT = 'rz_T - 25'
.param rz_w = '2u - 2*0.0148e-6'
.param rz_l = '2.75u - 2*3.85e-11'
.param rz_rsh = '1000*(1 + rz_rc)'
.param rz_tb = '1 - 9.39e-4*rz_dT + 2.51e-6*rz_dT*rz_dT'
.param rz_tt = '1 - 7.92e-3*rz_dT + 4.25e-5*rz_dT*rz_dT'
.param rz_val = 'rz_tb*(rz_l/rz_w)*rz_rsh + 2*rz_tt*85.45*(1u/rz_w)'

* Miller compensation with PDK poly resistor R_Z
CC Vo N006 {CCV}
RZ N005 N006 {rz_val}
* identical test resistor with 1 V across it, to read R_Z in each run
VT NT 0 1
RT NT 0 {rz_val}

* in .control: loop over the resistor corners inside the temperature loop
foreach tc -40 27 85 125
 foreach rc 0 0.2 -0.2
  alterparam rz_rc = $rc
  alterparam rz_T = $tc
  reset
  option temp = $tc
  save all @m.xm6.m0[gm]
  op
  let rz = -1/i(VT)
  let k = @m.xm6.m0[gm] * rz
  set s_rz = $&rz
  set s_k = $&k
  ...
  echo "$corner $tc rpdk_$rc $s_rz $s_k $&a0 $&ft $&pm $&f180 $&gm" >> ts_corners_results.txt
  destroy all
 end
end
```

B.4 Monte Carlo Set-Up (ngspice)

The Monte Carlo runs use ngspice-47 with the full GF180MCU library, which contains the statistical model sections. The PDK defines the transistors as subcircuits, so each device has an `X` prefix. Each run has 500 trials and uses a fixed random seed (`setseed 1`) so that it can be repeated. The listing below is the global-plus-mismatch run with the triode device M_Z; the mismatch-only run is identical except for `sw_stat_global=0` and the name of the results file. The 8× runs change only the bias-stack widths, to `WA=WB=10.75u` and `WP=43.84u`; M_Z and M_6 keep their original sizes.

```
* Two-stage op-amp, triode nulling device MZ: Monte Carlo of offset, A0, ft, PM
* Statistical switches: 0/1 = mismatch only, 1/0 = global only, 1/1 = realistic
.param sw_stat_global=1 sw_stat_mismatch=1
.param mc_skew=3 res_mc_skew=3 cap_mc_skew=3 fnoicor=0
.lib sm141064.ngspice statistical
.temp 27
.param E=0.44u
.param L0=0.56u
.param W1=6.85u W3=1.68u W5=13.7u W6=13.44u W7=54.8u W8=13.7u
.param WP=5.48u WA=1.344u WB=1.344u WZ=10.7u
.param CCV=350f

V1 VSS 0 0
V2 VDD 0 3.3

* ---- bias network ----
IREF N001 VSS 20u
XM8 N001 N001 VDD VDD pmos_3p3 w={W8} l={L0} ad={W8*E} as={W8*E} pd={2*W8+2*E} ps={2*W8+2*E}
XMP N002 N001 VDD VDD pmos_3p3 w={WP} l={L0} ad={WP*E} as={WP*E} pd={2*WP+2*E} ps={2*WP+2*E}
XMB N002 N002 N004 VSS nmos_3p3 w={WB} l={L0} ad={WB*E} as={WB*E} pd={2*WB+2*E} ps={2*WB+2*E}
XMA N004 N004 VSS VSS nmos_3p3 w={WA} l={L0} ad={WA*E} as={WA*E} pd={2*WA+2*E} ps={2*WA+2*E}

* ---- core (M1 = inverting, M2 = non-inverting input) ----
XM5 N003 N001 VDD VDD pmos_3p3 w={W5} l={L0} ad={W5*E} as={W5*E} pd={2*W5+2*E} ps={2*W5+2*E}
```

```
XM1 N007 vm N003 VDD pmos_3p3 w={W1} l={L0} ad={W1*E} as={W1*E} pd={2*W1+2*E} ps={2*W1+2*E}
XM2 N005 vp N003 VDD pmos_3p3 w={W1} l={L0} ad={W1*E} as={W1*E} pd={2*W1+2*E} ps={2*W1+2*E}
XM3 N007 N007 VSS VSS nmos_3p3 w={W3} l={L0} ad={W3*E} as={W3*E} pd={2*W3+2*E} ps={2*W3+2*E}
XM4 N005 N007 VSS VSS nmos_3p3 w={W3} l={L0} ad={W3*E} as={W3*E} pd={2*W3+2*E} ps={2*W3+2*E}
XM7 Vo N001 VDD VDD pmos_3p3 w={W7} l={L0} ad={W7*E} as={W7*E} pd={2*W7+2*E} ps={2*W7+2*E}
XM6 Vo N005 VSS VSS nmos_3p3 w={W6} l={L0} ad={W6*E} as={W6*E} pd={2*W6+2*E} ps={2*W6+2*E}
* Miller compensation with triode nulling device MZ
CC Vo N006 {CCV}
XMZ N005 N002 N006 VSS nmos_3p3 w={WZ} l={L0} ad={WZ*E} as={WZ*E} pd={2*WZ+2*E} ps={2*WZ+2*E}
CL Vo VSS 1p

* ---- bench: DC feedback through L, AC drive into v- through C ----
* DC: C open, L short -> unity-gain buffer, output at ~1.65 V
* AC: L open, C short -> vs drives v- directly, vo/vs = -A (open-loop gain)
Vcm vp 0 dc 1.65
LFB Vo vm 1e6
CFB vm vs 1e6
Vs vs 0 dc 0 ac 1

.control
set units = degrees
setseed 1
echo "rz_ohm k vos_V ptot_W a0_dB ft_Hz pm_deg" > ts_mz_mc_results.txt
repeat 500
  reset
  save all @m.xm6.m0[gm] @m.xmz.m0[gds]
  op
  * k = gm6 / gds(MZ) = gm6 * R_Z
  let k = @m.xm6.m0[gm] / @m.xmz.m0[gds]
  * on-resistance of MZ
  let rmz = 1/@m.xmz.m0[gds]
  set s_rmz = $&rmz
  set s_k = $&k
  let vos = v(vo) - v(vp)
  let ptot = -3.3*i(V2)
  set s_vos = $&vos
  set s_ptot = $&ptot
  ac dec 50 100 10g
  meas ac a0 max vdb(vo)
  meas ac ft when vdb(vo)=0
  let avec = -1*v(vo)
  let pha = ph(avec)
  meas ac ph find pha when vdb(vo)=0
  let pm = 180 + ph
  echo "$s_rmz $s_k $s_vos $s_ptot $&a0 $&ft $&pm" >> ts_mz_mc_results.txt
  destroy all
end
quit
.endc
.end
```

The PDK-resistor run replaces M_Z and its bias branch (MP, MB, MA) with the `ppolyf_u_1k` resistor ($2 \times 2.75\,\mu$m) and adds the PDK's resistor statistics. It uses the same seed and number of trials. ngspice-47 mis-converts the body resistor inside the PDK subcircuit, leaving only the contact resistance, so R_Z is written out from the PDK's own equations (body plus contacts, with the same temperature coefficients and statistical parameters). A test resistor with 1 V across it gives the value of R_Z in each trial. The lines that differ are:

```
* Two-stage op-amp, PDK poly resistor RZ: Monte Carlo of offset, A0, ft, PM
* (resistors vary only with sw_stat_global)
.lib sm141064.ngspice res_statistical

* ---- R_Z = GF180 ppolyf_u_1k, 2.75u x 2u, from the PDK's own equations ----
.param rz_T = 27
.param rz_dT = 'rz_T - 25'
.param rz_dw = '0.0148e-6*(1 + mc_dw_ppolyf_u_1k*res_mc_skew*sw_stat_global)'
.param rz_w = '2u - 2*rz_dw'
.param rz_l = '2.75u - 2*3.85e-11'
.param rz_rsh = 'rsh_ppolyf_u_1k*(1 + mc_rsh_ppolyf_u_1k/rsh_ppolyf_u_1k*res_mc_skew*sw_stat_global)'
.param rz_tb = '1 - 9.39e-4*rz_dT + 2.51e-6*rz_dT*rz_dT'
.param rz_body = 'rz_tb*(rz_l/rz_w)*rz_rsh'
```

```
.param rz_tt = '1 - 7.92e-3*rz_dT + 4.25e-5*rz_dT*rz_dT'
.param rz_cont = '2*rz_tt*(85.45 + mc_rt_ppolyf_u_1k*res_mc_skew*sw_stat_global)*(1u/rz_w)'
.param rz_val = 'rz_body + rz_cont'
RZ N005 N006 {rz_val}
* identical test resistor with 1 V across it, to read R_Z per trial
VT NT 0 1
RT NT 0 {rz_val}

* in .control: k = gm6 * R_Z
  save all @m.xm6.m0[gm]
  op
  let rz = -1/i(VT)
  let k = @m.xm6.m0[gm] * rz
```